



UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE E TECNOLOGIE

Corso di Laurea Magistrale in Fisica

Entangling Unravellings for Quantum State Engineering and Quantum Metrology

Tesi di laurea magistrale di:

Gabriele Campanello

Matricola 47116A

Relatore:

Prof. Marco Giovanni Genoni

Correlatore:

Dott. Eoin Anthony O'Connor

Anno accademico 2024-2025

A coloro che mi hanno insegnato a non mollare mai

True happiness is to enjoy the present, without anxious dependence upon the future, not to amuse ourselves with either hopes or fears but to rest satisfied with what we have, which is sufficient, for he that is so wants nothing. The greatest blessings of mankind are within us and within our reach. A wise man is content with his lot, whatever it may be, without wishing for what he has not.
- Seneca

Contents

1	Introduction to Quantum Mechanics	3
1.1	From classical physics to quantum mechanics	3
1.2	Postulates of quantum mechanics for closed and isolated systems	4
1.2.1	Postulate 1 : States	4
1.2.2	Postulate 2 : Observables	5
1.2.3	Postulate 3 : Measurements and conditional states	5
1.2.4	Postulate 4 : Dynamics	6
1.2.5	Postulate 5 : Composite systems	7
1.3	States and observables	7
1.3.1	Hilbert spaces and Dirac notation	7
1.3.2	Linear operators, outer products, and adjoints	8
1.3.3	Observables and spectral decomposition	10
1.4	Statistical ensembles and mixed states	11
1.4.1	Two-level systems and preferred bases	13
1.4.2	Bloch representation and Bloch sphere	13
1.5	Time evolution in quantum mechanics	14
1.5.1	Time-evolution operator and its general properties	15
1.5.2	Infinitesimal evolution and Hamiltonian generator	15
1.5.3	Time-dependent Hamiltonians and time ordering	17
1.6	Bipartite and multipartite systems	18
1.6.1	Tensor-product structure and product bases	18
1.6.2	Local operators, reduced states, and partial trace	19
1.6.3	Bell basis and parity sectors	19
1.6.4	Separable and entangled states: definition and methods	20
1.6.5	Concurrence for two-qubit states	23
1.6.6	Generic multipartite systems and collective spin observables	25
1.7	Spin squeezing	28
1.7.1	Collective spin operators and coherent spin states	29
1.7.2	Kitagawa-Ueda and Wineland definitions	29
1.7.3	Spin squeezing and symmetric states	31
2	Open Quantum Systems and continuously monitored systems	33
2.1	Open quantum systems: formalism	33
2.1.1	Generalized measurements, the POVMs	35
2.1.2	Quantum maps and completely positive transformations	37

2.1.3	Stinespring theorem and Kraus' operator-sum representation	38
-------	--	----

Bibliography		45
---------------------	--	-----------

Introduction

...

1

Introduction to Quantum Mechanics

1.1 From classical physics to quantum mechanics

By the end of the nineteenth century, classical physics seemed to offer a complete and convincing description of the natural world. Analytical mechanics, Maxwell's electrodynamics, and classical statistical physics formed a remarkably successful framework, capable of explaining a vast range of phenomena. Within this picture, the state of a physical system at a given time was assumed to determine, at least in principle, its future evolution. Observables were regarded as quantities possessing definite values, while the role of theory was to describe their behaviour through deterministic laws.

This confidence began to weaken when physicists turned their attention to microscopic phenomena. A number of experimental results, such as black-body radiation [1] and the photoelectric effect [2] could not be satisfactorily explained within the classical framework. These were not minor discrepancies that could be fixed by adjusting a few assumptions. Rather, they suggested that the classical picture itself was no longer adequate at the atomic scale.

The first attempts to deal with this crisis already required a significant break from classical intuition. Planck's introduction of energy quanta, Einstein's interpretation of the photoelectric effect, and Bohr's model of the hydrogen atom all pointed in the same direction. They represented an intermediate stage, in which classical concepts were still partially retained while new quantization rules were introduced in a rather phenomenological way [3].

A coherent formulation emerged only in the second half of the 1920s, with the development of matrix mechanics [4, 5] and wave mechanics [6]. Although these approaches looked very different at first, they were soon recognized as mathematically equivalent descriptions of the same physical theory. This marked the real birth of quantum mechanics as a general framework. From that point, the issue was no longer how to repair classical physics but to confirm that microscopic systems required a different language. In classical physics, the state of a system is ideally viewed as a complete specification of its physical properties while in quantum mechanics, this idea has to be revised. The state no longer provides, in general, definite values for all observables before a measurement is performed. Instead, it encodes the possible outcomes of a measurement and the statistical structure relating them.

In classical statistical physics, probabilities usually arise because one lacks complete in-

formation about the system. The probabilistic description is therefore associated with ignorance about an underlying state of affairs that is assumed to be well defined. It is totally different in quantum mechanics, where the probabilistic content is not merely a practical limitation of the observer but is built into the theory itself [7].

The conceptual consequences become even more striking when composite systems are considered. In the quantum framework, the state of a global system cannot always be reduced to independent states of its constituents. This gives rise to entanglement, one of the most distinctive features of the quantum realm. Quantum correlations are not simply stronger versions of classical correlations but they reflect a different logical structure, in which the whole cannot always be understood as a collection of independently defined parts.

What originally emerged from the study of atomic phenomena eventually became the foundation of several modern research areas. Quantum optics, many-body physics, quantum information, and quantum metrology all rely on genuinely non-classical features such as coherence, interference, and entanglement.

This perspective is especially relevant for the present thesis. The analysis of collective quantum properties in systems of many two-level atoms, and in particular the study of non-classical correlations and spin-squeezed states, belongs to the conceptual framework opened by quantum mechanics.

1.2 Postulates of quantum mechanics for closed and isolated systems

Having introduced the conceptual transition from classical physics to quantum mechanics, we now turn to the standard formulation of the theory [8, 9]. In its textbook form, quantum mechanics is first presented as a theory for closed and isolated systems. This setting is idealized, yet it provides the natural starting point to define the basic notions of state, observable, measurement, composition of systems, and time evolution.

This formulation is not the most general one. In particular, it is based on pure states, projective measurements, and unitary dynamics. These assumptions are fully adequate for introducing the theory in its simplest and most transparent form, but they will later need to be generalized when discussing realistic situations in which a system interacts with external degrees of freedom. For the purposes of the present chapter, however, the standard formulation is exactly the one we need. It allows us to establish the conceptual backbone of quantum mechanics before moving, in the next chapter, to the theory of open quantum systems.

1.2.1 Postulate 1 : States

To every closed physical system there corresponds a Hilbert space $\mathcal{H}$. The pure states of the system are represented by normalized vectors $|\psi\rangle \in \mathcal{H}$, namely

$$\langle\psi|\psi\rangle = 1. \tag{1.1}$$

More precisely, a physical pure state is not identified with a single vector, but with a ray in Hilbert space, since two vectors that differ only by an overall phase describe the same physical situation.

Indeed, if $|\psi\rangle \in \mathcal{H}$ is a normalized state vector, then for any real phase $\alpha \in \mathbb{R}$ the vector

$$|\psi'\rangle = e^{i\alpha} |\psi\rangle \quad (1.2)$$

represents the same physical pure state. This is the invariance of pure states under the action of the group $U(1)$ of global phase transformations.

The reason is that all measurable predictions remain unchanged under the replacement $|\psi\rangle \rightarrow e^{i\alpha} |\psi\rangle$.

An equivalent way to state the same fact is to observe that the rank-one projector associated with the pure state is invariant under a global phase transformation:

$$|\psi'\rangle\langle\psi'| = e^{i\alpha} |\psi\rangle\langle\psi| e^{-i\alpha} = |\psi\rangle\langle\psi|. \quad (1.3)$$

For this reason, pure states are properly identified with rays in Hilbert space, or equivalently with rank-one projectors.

If $|\psi_1\rangle$ and $|\psi_2\rangle$ are admissible states, then any normalized linear combination of them is again an admissible state. This introduces one of the most characteristic features of quantum theory, namely the superposition principle. This has no direct analogue in the classical description. It is the mathematical origin of interference phenomena and, more generally, of the non-classical structure of the theory. Quantum mechanics is insensitive to a common $U(1)$ phase multiplying the whole vector, but it is sensitive to relative phases between different components of a superposition.

1.2.2 Postulate 2 : Observables

Every observable quantity is associated with a self-adjoint operator acting on the Hilbert space of the system $\mathcal{H}$. If the observable is denoted by $\hat{X}$, its possible measurement outcomes are the eigenvalues appearing in its spectral decomposition. For the case of a discrete spectrum, one writes

$$\hat{X} = \sum_x x \hat{P}_x, \quad (1.4)$$

where the real numbers x are the possible outcomes and the operators $\hat{P}_x$ are the orthogonal projectors onto the corresponding eigenspaces.

This postulate establishes the link between physical quantities and operator theory: the fact that observables are represented by self-adjoint operators ensures, on one hand, that the possible outcomes are real, and on the other hand that the spectral theorem may be used to connect the formal object $\hat{X}$ with experimentally accessible values. For this reason, even when one speaks of the measurement of an observable, the truly relevant objects are not only its eigenvalues, but also the corresponding projectors.

1.2.3 Postulate 3 : Measurements and conditional states

If the system is prepared in the pure state $|\psi\rangle$ and one measures the observable $\hat{X}$, quantum mechanics does not in general predict a single certain result. Instead, it assigns

probabilities to the possible outcomes. For a discrete observable, the probability of obtaining the value x is given by the Born rule

$$p(x) = \langle \psi | \hat{P}_x | \psi \rangle. \quad (1.5)$$

Correspondingly, the expectation value of the observable reads

$$\langle \hat{X} \rangle = \langle \psi | \hat{X} | \psi \rangle. \quad (1.6)$$

The Born rule is one of the central ingredients of the entire formalism. It is the rule that connects the mathematical description of the state with the statistical predictions of the theory. Once the state and the observable are assigned, the Born rule determines the probability distribution of the measurement outcomes. In this sense, it is the point at which the formalism becomes experimentally meaningful.

The standard formulation also includes a rule for the state change induced by an ideal projective measurement. If the outcome x is obtained, the state immediately after the measurement is given by

$$|\psi\rangle \longrightarrow |\psi_x\rangle = \frac{\hat{P}_x |\psi\rangle}{\sqrt{\langle \psi | \hat{P}_x | \psi \rangle}}. \quad (1.7)$$

It expresses the fact that the state after the measurement is the normalized projection of the initial state onto the eigenspace corresponding to the observed result.

1.2.4 Postulate 4 : Dynamics

In the absence of noise and measurements, the evolution of a closed and isolated quantum system is unitary. If the state of the system at time t_0 is $|\psi(t_0)\rangle$, then at time t it is given by

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle, \quad (1.8)$$

where $\hat{U}(t, t_0)$ is a unitary operator, namely

$$\hat{U}^\dagger(t, t_0) \hat{U}(t, t_0) = \hat{U}(t, t_0) \hat{U}^\dagger(t, t_0) = \mathbb{1}. \quad (1.9)$$

If the dynamics is generated by a Hamiltonian $\hat{H}$, the same evolution may be written in differential form through the Schrödinger equation

$$i \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle. \quad (1.10)$$

Once the initial state and the Hamiltonian are given, the state at any later time is uniquely determined. In this respect, quantum mechanics preserves a form of deterministic evolution, but only between measurements. The probabilistic character of the theory enters when one connects the evolving state to the outcomes of observations through the Born rule.

It is also worth emphasizing that unitarity guarantees that the normalization of the state is preserved in time, and thus that total probability remains equal to one throughout the evolution of an isolated system.

1.2.5 Postulate 5 : Composite systems

If a physical system is composed of two subsystems, described respectively by the Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, then the composite system is described on the tensor-product space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \quad (1.11)$$

More generally, a multipartite system is described by the tensor product of the Hilbert spaces associated with all its constituents.

This postulate is of fundamental importance, it is precisely this structure that makes entanglement possible. Indeed, not every state in the global Hilbert space can be written as a product of states of the individual subsystems. Therefore, the description of the whole cannot in general be reduced to independent descriptions of its parts.

In the classical picture, correlations may be interpreted as reflecting incomplete information about local properties that each subsystem already possesses. In the quantum case, the tensor-product structure allows for correlations that are genuinely non-classical. For this reason, the postulate on composite systems is not just a formal extension of the single-system case. It is the starting point for the entire theory of quantum correlations. For the whole thesis, this postulate will be important. Since our interest lies in collective properties of many two-level systems, all the relevant notions of multipartite state, collective observable, and entanglement rely directly on the tensor-product description.

These assumptions are entirely appropriate for a first formulation of the theory and will be sufficient for the developments of the present chapter. However, they are not general enough to describe realistic situations in which the system interacts with an external environment, or in which measurements are not ideal projective measurements. For this reason, they should be regarded here as the natural starting point of the discussion. In Chapter 2, we will move to a more general formulation based on density operators, generalized measurements, and open-system dynamics.

1.3 States and observables

The postulates introduced in the previous section provide the conceptual skeleton of quantum mechanics. We now discuss in more detail the mathematical objects entering that formulation. In particular, we clarify the notions of physical system, quantum state, and observable, and we introduce the Dirac notation that will be used throughout the thesis [9, 10].

1.3.1 Hilbert spaces and Dirac notation

In the standard formulation adopted in this chapter, a physical system is associated with a Hilbert space $\mathcal{H}$. For the purposes of the present discussion, and consistently with the framework that will be used in the following sections, we mainly refer to finite-dimensional Hilbert spaces and to observables with discrete spectrum. This is the natural setting for the description of qubits and, more generally, of finite collections of two-level systems.

A Hilbert space is a complex vector space endowed with an inner product. Therefore, if $|\psi\rangle, |\phi\rangle \in \mathcal{H}$ and $\alpha, \beta \in \mathbb{C}$, then $\alpha|\psi\rangle + \beta|\phi\rangle \in \mathcal{H}$.

We adopt the Dirac notation throughout. A vector in $\mathcal{H}$ is denoted by a ket $|\psi\rangle$ and the corresponding dual vector is denoted by a bra $\langle\psi| = (|\psi\rangle)^\dagger$. The inner product between two vectors $|\phi\rangle$ and $|\psi\rangle$ is written as $\langle\phi|\psi\rangle \in \mathbb{C}$. In the physics convention, the inner product is linear in the second entry and antilinear in the first one, namely

$$\langle\phi|\alpha\psi + \beta\chi\rangle = \alpha\langle\phi|\psi\rangle + \beta\langle\phi|\chi\rangle, \quad (1.12)$$

while

$$\langle\alpha\phi + \beta\chi|\psi\rangle = \alpha^*\langle\phi|\psi\rangle + \beta^*\langle\chi|\psi\rangle. \quad (1.13)$$

Moreover, one has $\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*$ and $\langle\psi|\psi\rangle \geq 0$, with equality if and only if $|\psi\rangle = 0$. The norm induced by the inner product is $\|\psi\| = \sqrt{\langle\psi|\psi\rangle}$. Physical pure states are represented by normalized vectors, namely vectors satisfying $\langle\psi|\psi\rangle = 1$. As already discussed, the physical content does not change under multiplication by a global phase. Hence, the pure state is properly associated with the ray generated by $|\psi\rangle$ rather than with the vector itself.

Let now $\{|e_n\rangle\}_{n=1}^d$ be an orthonormal basis of $\mathcal{H}$. By orthonormality,

$$\langle e_m|e_n\rangle = \delta_{mn}, \quad (1.14)$$

where δ_{mn} is the Kronecker delta. The completeness of the basis is expressed by the resolution of the identity

$$\sum_{n=1}^d |e_n\rangle\langle e_n| = \mathbb{1}. \quad (1.15)$$

Using Eq. (1.15), any state vector may be expanded as

$$|\psi\rangle = \sum_{n=1}^d |e_n\rangle \langle e_n|\psi\rangle = \sum_{n=1}^d c_n |e_n\rangle, \quad (1.16)$$

where the expansion coefficients are $c_n = \langle e_n|\psi\rangle$, and therefore $\langle\psi| = \sum_{n=1}^d c_n^* \langle e_n|$. If $|\psi\rangle$ is normalized, then Eqs. (1.14) and (1.16) imply

$$1 = \langle\psi|\psi\rangle = \sum_{n=1}^d |c_n|^2. \quad (1.17)$$

This expression already shows the probabilistic interpretation of the coefficients once a basis has been fixed. In particular, when the chosen basis is the eigenbasis of an observable, the moduli squared of the coefficients become the probabilities of the corresponding measurement outcomes through the Born rule.

1.3.2 Linear operators, outer products, and adjoints

Besides vectors, quantum mechanics makes use of linear operators acting on $\mathcal{H}$. If $\hat{A}$ is a linear operator, then $\hat{A}(\alpha|\psi\rangle + \beta|\phi\rangle) = \alpha\hat{A}|\psi\rangle + \beta\hat{A}|\phi\rangle$. Given two vectors $|\phi\rangle$ and $|\psi\rangle$,

one may construct the outer product $|\phi\rangle\langle\psi|$, which is itself a linear operator and acts on an arbitrary vector $|\chi\rangle$ as

$$(|\phi\rangle\langle\psi|)|\chi\rangle = |\phi\rangle\langle\psi|\chi\rangle. \quad (1.18)$$

In particular, the operator $|\psi\rangle\langle\psi|$ is the rank-one projector associated with the pure state $|\psi\rangle$. For a normalized state, it satisfies $(|\psi\rangle\langle\psi|)^2 = |\psi\rangle\langle\psi|$ and $\text{Tr}(|\psi\rangle\langle\psi|) = 1$.

This operator representation will become especially useful in the next chapter, where the density-operator formalism will be introduced in full generality.

Let us now consider the matrix representation of an operator. Once a basis $\{|e_n\rangle\}_{n=1}^d$ is fixed, the matrix elements of $\hat{A}$ are defined as $A_{mn} = \langle e_m|\hat{A}|e_n\rangle$. Using the completeness relation twice, the operator may be reconstructed as

$$\hat{A} = \sum_{m,n=1}^d A_{mn} |e_m\rangle\langle e_n|. \quad (1.19)$$

Therefore, once a basis is chosen, vectors and operators admit the familiar column-vector and matrix representations of linear algebra. Dirac notation is simply a more flexible and basis-independent way of encoding the same information.

The adjoint of an operator $\hat{A}$ is defined by the relation

$$\langle\phi|\hat{A}|\psi\rangle = \langle\hat{A}^\dagger\phi|\psi\rangle, \quad (1.20)$$

or equivalently by

$$\langle\phi|\hat{A}|\psi\rangle = \langle\psi|\hat{A}^\dagger|\phi\rangle^*. \quad (1.21)$$

An operator is said to be self-adjoint if

$$\hat{A}^\dagger = \hat{A}. \quad (1.22)$$

In the finite-dimensional setting considered in this chapter, the terms self-adjoint and Hermitian may be used interchangeably.

Self-adjoint operators play a distinguished role in quantum mechanics because observables are represented precisely by operators of this kind. A first important property is that their expectation values are always real. Indeed, if $\hat{A}^\dagger = \hat{A}$, then

$$\langle\psi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\psi\rangle = \langle\psi|\hat{A}|\psi\rangle, \quad (1.23)$$

hence $\langle\psi|\hat{A}|\psi\rangle \in \mathbb{R}$.

A second key property is that the eigenvalues of a self-adjoint operator are real. If $\hat{A}|a\rangle = a|a\rangle$, then

$$a\langle a|a\rangle = \langle a|\hat{A}|a\rangle = \langle a|\hat{A}|a\rangle^* = a^*\langle a|a\rangle, \quad (1.24)$$

and therefore $a = a^*$.

Finally, eigenvectors corresponding to distinct eigenvalues are orthogonal. If $\hat{A}|a\rangle = a|a\rangle$ and $\hat{A}|b\rangle = b|b\rangle$ with $a \neq b$, then

$$a\langle a|b\rangle = \langle\hat{A}a|b\rangle = \langle a|\hat{A}b\rangle = b\langle a|b\rangle, \quad (1.25)$$

so that $(a - b) \langle a|b \rangle = 0$, namely $\langle a|b \rangle = 0$.

In finite dimension, these properties are summarized by the spectral theorem: every self-adjoint operator is diagonalizable in an orthonormal basis and may be written as

$$\hat{A} = \sum_a a \hat{P}_a, \quad (1.26)$$

where the coefficients $a \in \mathbb{R}$ are the eigenvalues and the operators $\hat{P}_a$ are the orthogonal projectors onto the corresponding eigenspaces. In the nondegenerate case, one simply has

$$\hat{A} = \sum_a a |a\rangle\langle a|. \quad (1.27)$$

For this reason, self-adjoint operators provide exactly the mathematical structure needed to represent observables.

1.3.3 Observables and spectral decomposition

We are now in the position to discuss observables more concretely. Let $\hat{X}$ be a self-adjoint operator acting on $\mathcal{H}$. In the finite-dimensional case, the spectral theorem guarantees that $\hat{X}$ may be diagonalized and written in the form

$$\hat{X} = \sum_x x \hat{P}_x, \quad (1.28)$$

where $x \in \mathbb{R}$ are the eigenvalues of the observable and $\hat{P}_x$ are the orthogonal projectors onto the corresponding eigenspaces. These projectors satisfy $\hat{P}_x \hat{P}_{x'} = \delta_{xx'} \hat{P}_x$ and $\sum_x \hat{P}_x = \mathbf{1}$. If the spectrum is nondegenerate, each eigenspace is one-dimensional and one may write

$$\hat{P}_x = |x\rangle\langle x|, \quad \hat{X} = \sum_x x |x\rangle\langle x|, \quad (1.29)$$

with $\hat{X}|x\rangle = x|x\rangle$.

The expectation value of $\hat{X}$ in the pure state $|\psi\rangle$ is

$$\langle \hat{X} \rangle_\psi = \langle \psi | \hat{X} | \psi \rangle. \quad (1.30)$$

By inserting the spectral decomposition (1.28), one finds

$$\langle \hat{X} \rangle_\psi = \sum_x x \langle \psi | \hat{P}_x | \psi \rangle. \quad (1.31)$$

The coefficients $p(x) = \langle \psi | \hat{P}_x | \psi \rangle$ are non-negative and sum to one, and therefore define a probability distribution over the possible outcomes. This is the precise way in which the state vector determines the statistics of the observable.

A second quantity of central importance is the variance of $\hat{X}$ in the state $|\psi\rangle$,

$$\left(\Delta^2 \hat{X} \right)_\psi = \langle \hat{X}^2 \rangle_\psi - \langle \hat{X} \rangle_\psi^2. \quad (1.32)$$

The corresponding standard deviation is simply $\left(\Delta\hat{X}\right)_\psi = \sqrt{\left(\Delta^2\hat{X}\right)_\psi}$. These quantities characterize the spread of the outcomes around the mean value and will become particularly relevant later on, when discussing collective observables and the reduction of quantum fluctuations below suitable reference levels.

It is also useful to comment briefly on compatibility of observables. If two observables $\hat{X}$ and $\hat{Y}$ commute, namely

$$[\hat{X}, \hat{Y}] = 0, \quad (1.33)$$

then, in the finite-dimensional setting, they admit a common eigenbasis, up to the usual care required in the presence of degeneracies. In that case, the two observables may be simultaneously diagonalized and can be jointly sharp in the same states. When the commutator does not vanish, such a common diagonalization is in general not possible. This is the algebraic origin of quantum incompatibility and, more generally, of the fact that different experimental arrangements may probe genuinely different aspects of the same physical system.

1.4 Statistical ensembles and mixed states

The vector-state formalism is fully adequate when the preparation procedure singles out one definite pure state. In that case, the state of the system is completely described by a normalized vector in Hilbert space, or equivalently by the associated rank-one projector. In many physically relevant situations, however, this description is no longer sufficient. The preparation may be only partially controlled, or it may produce different pure states with given classical probabilities. One is then naturally led to consider statistical ensembles rather than individual state vectors.

More precisely, suppose that a preparation device produces the pure state $|\psi_j\rangle$ with probability p_j , where $p_j \geq 0$ and $\sum_j p_j = 1$. One then speaks of a statistical ensemble of pure states,

$$\mathcal{E} = \{p_j, |\psi_j\rangle\}_j. \quad (1.34)$$

The physical meaning of this description is : before any measurement is performed, one does not know which of the vectors $|\psi_j\rangle$ has actually been prepared in a given run, but one does know the probabilities with which the different alternatives occurs.

If $\hat{X}$ is an observable, the expectation value associated with the ensemble is obtained by averaging over the possible preparations:

$$\langle \hat{X} \rangle = \sum_j p_j \langle \psi_j | \hat{X} | \psi_j \rangle. \quad (1.35)$$

At this stage, an important structural fact emerges. Although the ensemble is defined as a collection of vectors and probabilities, all measurable predictions depend on it only through a single operator. Indeed, by linearity one may rewrite Eq. (1.35) as

$$\langle \hat{X} \rangle = \text{Tr}(\hat{\rho}\hat{X}), \quad (1.36)$$

provided one introduces the density operator

$$\hat{\rho} = \sum_j p_j |\psi_j\rangle\langle\psi_j|. \quad (1.37)$$

This is the natural generalization of the projector associated with a pure state. In the special case in which one of the probabilities is equal to one, say $p_{j_0} = 1$, Eq. (1.37) reduces to $\hat{\rho} = |\psi_{j_0}\rangle\langle\psi_{j_0}|$, and the pure-state formalism is recovered.

The density operator is therefore not introduced as an abstract mathematical object, but as the unique operator encoding all statistical predictions associated with a preparation procedure. In particular, if two distinct ensembles define the same operator $\hat{\rho}$, then they yield the same expectation values for every observable and are therefore physically indistinguishable as far as measurements on the system are concerned.

The operator $\hat{\rho}$ satisfies three basic properties. First, it is self-adjoint. Second, it is positive. Third, it has unit trace. Explicitly,

$$\hat{\rho}^\dagger = \hat{\rho}, \quad \langle\phi|\hat{\rho}|\phi\rangle \geq 0 \quad \forall |\phi\rangle \in \mathcal{H}, \quad \text{Tr}(\hat{\rho}) = 1. \quad (1.38)$$

The proof follows immediately from Eq. (1.37). Self-adjointness is obvious, while positivity follows from

$$\langle\phi|\hat{\rho}|\phi\rangle = \sum_j p_j \langle\phi||\psi_j\rangle\langle\psi_j||\phi\rangle = \sum_j p_j |\langle\psi_j|\phi\rangle|^2 \geq 0. \quad (1.39)$$

Finally,

$$\text{Tr}(\hat{\rho}) = \sum_j p_j \text{Tr}(|\psi_j\rangle\langle\psi_j|) = \sum_j p_j = 1. \quad (1.40)$$

Conversely, every positive trace-one operator defines a quantum state. In this sense, the density-operator formalism is not an optional extension of the vector formalism, but its proper completion. Pure states appear as a special case, while mixed states arise whenever the preparation cannot be associated with a single ray in Hilbert space.

The set of quantum states has a convex structure. If $\hat{\rho}_1$ and $\hat{\rho}_2$ are density operators and $0 \leq p \leq 1$, then

$$\hat{\rho} = p\hat{\rho}_1 + (1-p)\hat{\rho}_2 \quad (1.41)$$

is again a density operator. This reflects the elementary fact that one may probabilistically mix two preparation procedures. Pure states are precisely the extremal points of this convex set.

If the system is in the pure state $|\psi\rangle$, then $\hat{\rho} = |\psi\rangle\langle\psi|$ and therefore $\hat{\rho}^2 = \hat{\rho}$. More generally,

$$\hat{\rho} \text{ pure} \iff \hat{\rho}^2 = \hat{\rho} \iff \text{Tr}(\hat{\rho}^2) = 1. \quad (1.42)$$

If instead $\text{Tr}(\hat{\rho}^2) < 1$, the state is mixed. The quantity

$$\gamma(\hat{\rho}) = \text{Tr}(\hat{\rho}^2) \quad (1.43)$$

is called the purity. It provides a simple measure of the degree of mixedness of the state.

A key conceptual point is that the ensemble decomposition of a mixed state is, in general, not unique. Different ensembles may define the same density operator and therefore the same statistical predictions for all observables. For a qubit, for instance,

$$\frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1| = \frac{1}{2} |+\rangle\langle +| + \frac{1}{2} |-\rangle\langle -| = \frac{\mathbb{1}}{2}, \quad (1.44)$$

where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. The two ensembles correspond to different preparation procedures, but they define the same statistical operator. Hence, from the point of view of measurement statistics, they represent the same state.

This observation is of fundamental importance. It shows that the density operator is the physically relevant object, whereas the ensemble decomposition is not unique and should not be regarded as part of the intrinsic definition of the state. This point will become even more important in the next chapter, where mixed states will arise not only from classical uncertainty in the preparation, but also from the reduced description of subsystems entangled with external degrees of freedom.

1.4.1 Two-level systems and preferred bases

We now specialize to a two-level system, or qubit. Its Hilbert space is $\mathbb{C}^2$. We choose as computational basis the orthonormal basis

$$\mathcal{B}_z = \{|0\rangle, |1\rangle\}, \quad \langle 0|0\rangle = \langle 1|1\rangle = 1, \quad \langle 0|1\rangle = 0. \quad (1.45)$$

With the convention adopted throughout, $\sigma_z |0\rangle = |0\rangle$ and $\sigma_z |1\rangle = -|1\rangle$, so that, in the ordered basis $\{|0\rangle, |1\rangle\}$,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (1.46)$$

The states introduced above form the eigenbasis of σ_x , namely $\sigma_x |\pm\rangle = \pm |\pm\rangle$. We therefore denote $\mathcal{B}_x = \{|+\rangle, |-\rangle\}$ as the σ_x eigenbasis.

A generic pure qubit state may be written as

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (1.47)$$

Up to a global phase, it can always be parametrized as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle, \quad (1.48)$$

with $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. The corresponding pure-state density operator is simply $\hat{\rho}_\psi = |\psi\rangle\langle\psi|$.

1.4.2 Bloch representation and Bloch sphere

Since $\{\mathbb{1}, \sigma_x, \sigma_y, \sigma_z\}$ is a basis for the space of 2×2 matrices, every qubit density operator admits the decomposition

$$\hat{\rho} = \frac{1}{2} (\mathbb{1} + \vec{r} \cdot \vec{\sigma}), \quad (1.49)$$

where $\vec{r} = (r_x, r_y, r_z) \in \mathbb{R}^3$ is the Bloch vector and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. Explicitly,

$$\hat{\rho} = \frac{1}{2} \begin{pmatrix} 1 + r_z & r_x - ir_y \\ r_x + ir_y & 1 - r_z \end{pmatrix}. \quad (1.50)$$

The components of $\vec{r}$ are obtained as expectation values of the Pauli operators,

$$r_k = \text{Tr}(\hat{\rho}\sigma_k), \quad k \in \{x, y, z\}, \quad (1.51)$$

since $\text{Tr}(\sigma_k) = 0$ and $\text{Tr}(\sigma_j\sigma_k) = 2\delta_{jk}$.

For the pure state in eq. (1.48), one finds

$$r_x = \sin \theta \cos \phi, \quad r_y = \sin \theta \sin \phi, \quad r_z = \cos \theta. \quad (1.52)$$

Thus, pure qubit states are in one-to-one correspondence with points on the unit sphere in $\mathbb{R}^3$, the Bloch sphere.

The positivity of $\hat{\rho}$ imposes a restriction on $\vec{r}$. Using $(\vec{r} \cdot \vec{\sigma})^2 = \|\vec{r}\|^2 \mathbb{1}$, one finds that the eigenvalues of $\hat{\rho}$ are

$$\lambda_{\pm} = \frac{1 \pm \|\vec{r}\|}{2}. \quad (1.53)$$

Therefore,

$$\hat{\rho} \geq 0 \iff \|\vec{r}\| \leq 1. \quad (1.54)$$

This means that the physically allowed states fill the Bloch ball. Pure states lie on its surface, $\|\vec{r}\| = 1$, whereas mixed states lie in its interior, $\|\vec{r}\| < 1$. The maximally mixed state is the center of the ball, namely $\hat{\rho}_{\text{mm}} = \mathbb{1}/2$ and $\vec{r} = 0$.

The relation between the Bloch vector and the purity is immediate:

$$\text{Tr}(\hat{\rho}^2) = \frac{1 + \|\vec{r}\|^2}{2}. \quad (1.55)$$

Hence,

$$\text{Tr}(\hat{\rho}^2) = 1 \iff \|\vec{r}\| = 1, \quad \text{Tr}(\hat{\rho}^2) < 1 \iff \|\vec{r}\| < 1. \quad (1.56)$$

Thus, the length of the Bloch vector directly quantifies the degree of purity of the state.

Finally, any qubit observable may be written as

$$\hat{X} = a_0 \mathbb{1} + \vec{a} \cdot \vec{\sigma}, \quad (1.57)$$

with $a_0 \in \mathbb{R}$ and $\vec{a} \in \mathbb{R}^3$. Its expectation value in the state $\hat{\rho}$ is

$$\langle \hat{X} \rangle = \text{Tr}(\hat{\rho}\hat{X}) = a_0 + \vec{a} \cdot \vec{r}. \quad (1.58)$$

In particular, one has $\langle \sigma_x \rangle = r_x$, $\langle \sigma_y \rangle = r_y$, and $\langle \sigma_z \rangle = r_z$.

1.5 Time evolution in quantum mechanics

We now discuss in more detail the dynamical postulate introduced in the previous section. For a closed and isolated quantum system, the time evolution is described by a family of unitary operators acting on the Hilbert space of the system. This is the natural quantum analogue of reversible evolution in the absence of external interactions.

1.5.1 Time-evolution operator and its general properties

Let $|\psi(t_0)\rangle$ be the state of the system at time t_0 . Its state at time t is written as

$$|\psi(t)\rangle = \hat{U}(t, t_0) |\psi(t_0)\rangle, \quad (1.59)$$

where $\hat{U}(t, t_0)$ is the time-evolution operator from t_0 to t . Since the evolution is unitary, one has

$$\hat{U}^\dagger(t, t_0) \hat{U}(t, t_0) = \hat{U}(t, t_0) \hat{U}^\dagger(t, t_0) = \mathbb{1}. \quad (1.60)$$

As a consequence, the norm of the state is preserved:

$$\langle \psi(t) | \psi(t) \rangle = \langle \psi(t_0) | \hat{U}^\dagger(t, t_0) \hat{U}(t, t_0) | \psi(t_0) \rangle = \langle \psi(t_0) | \psi(t_0) \rangle. \quad (1.61)$$

More generally,

$$\langle \phi(t) | \psi(t) \rangle = \langle \phi(t_0) | \hat{U}^\dagger(t, t_0) \hat{U}(t, t_0) | \psi(t_0) \rangle = \langle \phi(t_0) | \psi(t_0) \rangle. \quad (1.62)$$

Hence, transition probabilities are invariant under the evolution.

The propagator $\hat{U}(t, t_0)$ must satisfy a consistency condition. If $t_0 \leq t_1 \leq t_2$, evolving first from t_0 to t_1 and then from t_1 to t_2 must be equivalent to evolving directly from t_0 to t_2 . Therefore,

$$\hat{U}(t_2, t_1) \hat{U}(t_1, t_0) = \hat{U}(t_2, t_0). \quad (1.63)$$

Moreover, $\hat{U}(t_0, t_0) = \mathbb{1}$ and, by combining Eqs. (1.60) and (1.63), one obtains $\hat{U}^{-1}(t, t_0) = \hat{U}(t_0, t) = \hat{U}^\dagger(t, t_0)$.

1.5.2 Infinitesimal evolution and Hamiltonian generator

The local structure of the dynamics is obtained by considering an infinitesimal time step. For a sufficiently regular propagator, one may write

$$\hat{U}(t + dt, t) = \mathbb{1} - i \hat{G}(t) dt + \mathcal{O}(dt^2), \quad (1.64)$$

where $\hat{G}(t)$ is the generator of the evolution. Unitarity imposes a constraint on $\hat{G}(t)$.

Indeed, from

$$\hat{U}^\dagger(t + dt, t) \hat{U}(t + dt, t) = \mathbb{1} \quad (1.65)$$

and Eq. (1.64), one finds

$$\left(\mathbb{1} + i \hat{G}^\dagger(t) dt \right) \left(\mathbb{1} - i \hat{G}(t) dt \right) = \mathbb{1} + i \left[\hat{G}^\dagger(t) - \hat{G}(t) \right] dt + \mathcal{O}(dt^2). \quad (1.66)$$

Therefore,

$$\hat{G}^\dagger(t) = \hat{G}(t). \quad (1.67)$$

The generator of a unitary evolution must thus be self-adjoint. It is therefore identified with the Hamiltonian of the system [11], namely $\hat{G}(t) = \hat{H}(t)$.

The same result may be written in a more global form. By differentiating the identity

$$\hat{U}(t + dt, t_0) = \hat{U}(t + dt, t) \hat{U}(t, t_0) \quad (1.68)$$

and using Eq. (1.64), one obtains

$$\hat{U}(t + dt, t_0) = \left(\mathbb{1} - i \hat{H}(t) dt \right) \hat{U}(t, t_0) + \mathcal{O}(dt^2). \quad (1.69)$$

Subtracting $\hat{U}(t, t_0)$ from both sides, dividing by dt , and taking the limit $dt \rightarrow 0$, one finds

$$i \frac{\partial}{\partial t} \hat{U}(t, t_0) = \hat{H}(t) \hat{U}(t, t_0), \quad (1.70)$$

with initial condition

$$\hat{U}(t_0, t_0) = \mathbb{1}. \quad (1.71)$$

This is the evolution equation for the time-evolution operator. By applying it to the initial state $|\psi(t_0)\rangle$, one immediately obtains the Schrödinger equation

$$i \frac{d}{dt} |\psi(t)\rangle = \hat{H}(t) |\psi(t)\rangle. \quad (1.72)$$

When the Hamiltonian is time independent, namely $\hat{H}(t) = \hat{H}$, the propagator depends only on the time difference, $\hat{U}(t, t_0) = \hat{U}(t - t_0)$. By introducing $\tau = t - t_0$, one has

$$\hat{U}(\tau_1) \hat{U}(\tau_2) = \hat{U}(\tau_1 + \tau_2), \quad (1.73)$$

together with $\hat{U}(0) = \mathbb{1}$. Hence, the time evolution defines a one-parameter unitary group.

At this point one may invoke Stone's theorem [12]. In the present setting, the theorem states that every strongly continuous one-parameter unitary group is generated by a unique self-adjoint operator $\hat{H}$ such that

$$\hat{U}(\tau) = e^{-i\hat{H}\tau}. \quad (1.74)$$

Conversely, every self-adjoint operator $\hat{H}$ defines a strongly continuous one-parameter unitary group through Eq. (1.74). This gives a precise mathematical meaning to the statement that the Hamiltonian is the generator of time translations.

Equation (1.74) is written in units such that $\hbar = 1$. If $\hbar$ is kept explicit, one has

$$\hat{U}(\tau) = e^{-i\hat{H}\tau/\hbar}. \quad (1.75)$$

Throughout the following we adopt the natural units: unless otherwise stated, we set $\hbar = 1$.

The self-adjointness of $\hat{H}$ guarantees the unitarity of the evolution, since $\hat{U}^\dagger(\tau) = \left(e^{-i\hat{H}\tau} \right)^\dagger = e^{i\hat{H}\tau}$ and therefore $\hat{U}^\dagger(\tau) \hat{U}(\tau) = \mathbb{1}$. In the time-independent case, Eq. (1.70) can be solved explicitly:

$$\hat{U}(t, t_0) = e^{-i\hat{H}(t-t_0)}, \quad (1.76)$$

so that

$$|\psi(t)\rangle = e^{-i\hat{H}(t-t_0)} |\psi(t_0)\rangle. \quad (1.77)$$

1.5.3 Time-dependent Hamiltonians and time ordering

The time-dependent case is slightly more delicate. Formally, Eq. (1.70) suggests the expression

$$\hat{U}(t, t_0) \stackrel{?}{=} \exp \left[-i \int_{t_0}^t \hat{H}(s) ds \right]. \quad (1.78)$$

However, this formula is valid only when

$$[\hat{H}(t_1), \hat{H}(t_2)] = 0 \quad \forall t_1, t_2. \quad (1.79)$$

Indeed, if Hamiltonians at different times do not commute, the exponential of the time integral is not sufficient to reproduce the correct ordered product of infinitesimal propagators.

In the general case, the formal solution is written as the time-ordered exponential

$$\hat{U}(t, t_0) = \mathcal{T} \exp \left[-i \int_{t_0}^t \hat{H}(s) ds \right], \quad (1.80)$$

where $\mathcal{T}$ denotes chronological ordering. More precisely, for a product of Hamiltonians evaluated at different times, the action of $\mathcal{T}$ is defined by arranging the operators from left to right in decreasing time order. For instance,

$$\mathcal{T}[\hat{H}(t_1)\hat{H}(t_2)] = \begin{cases} \hat{H}(t_1)\hat{H}(t_2), & t_1 > t_2, \\ \hat{H}(t_2)\hat{H}(t_1), & t_2 > t_1, \end{cases} \quad (1.81)$$

and, more generally,

$$\mathcal{T}[\hat{H}(t_1)\hat{H}(t_2) \cdots \hat{H}(t_n)] = \hat{H}(t_{\pi(1)})\hat{H}(t_{\pi(2)}) \cdots \hat{H}(t_{\pi(n)}), \quad (1.82)$$

where the permutation π is such that

$$t_{\pi(1)} \geq t_{\pi(2)} \geq \cdots \geq t_{\pi(n)}. \quad (1.83)$$

Equation (1.80) may be expanded as the Dyson series [13],

$$\hat{U}(t, t_0) = \mathbb{1} + \sum_{n=1}^{\infty} (-i)^n \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n \hat{H}(t_1)\hat{H}(t_2) \cdots \hat{H}(t_n). \quad (1.84)$$

The first terms are therefore

$$\hat{U}(t, t_0) = \mathbb{1} - i \int_{t_0}^t \hat{H}(t_1) dt_1 + (-i)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 \hat{H}(t_1)\hat{H}(t_2) + \cdots. \quad (1.85)$$

This ordered structure ensures that the solution correctly reproduces the composition of infinitesimal propagators even when the Hamiltonian at different times does not commute with itself.

1.6 Bipartite and multipartite systems

We now discuss in more detail the structure of composite quantum systems. This is the natural setting in which quantum correlations, and in particular entanglement, arise. Since the present thesis is ultimately concerned with collective properties of many two-level systems, the tensor-product structure of the Hilbert space plays a central role from the very beginning.

1.6.1 Tensor-product structure and product bases

According to the composition postulate of quantum mechanics, if two subsystems A and B are associated with Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, then the composite system is described on the tensor-product space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \quad (1.86)$$

If $\{|a_i\rangle\}_{i=1}^{d_A}$ and $\{|b_j\rangle\}_{j=1}^{d_B}$ are orthonormal bases of $\mathcal{H}_A$ and $\mathcal{H}_B$, then the set

$$\mathcal{B}_{AB} = \{|a_i\rangle \otimes |b_j\rangle\}_{i,j} \quad (1.87)$$

is an orthonormal basis of $\mathcal{H}_{AB}$, since

$$(\langle a_i| \otimes \langle b_j|)(|a_{i'}\rangle \otimes |b_{j'}\rangle) = \langle a_i|a_{i'}\rangle \langle b_j|b_{j'}\rangle = \delta_{ii'}\delta_{jj'}. \quad (1.88)$$

Therefore, any pure bipartite state may be expanded as

$$|\Psi\rangle_{AB} = \sum_{i=1}^{d_A} \sum_{j=1}^{d_B} c_{ij} |a_i\rangle \otimes |b_j\rangle, \quad (1.89)$$

with normalization condition

$$\sum_{i,j} |c_{ij}|^2 = 1. \quad (1.90)$$

Operators on the tensor-product space are defined accordingly. If $\hat{A}$ acts on $\mathcal{H}_A$ and $\hat{B}$ acts on $\mathcal{H}_B$, then $\hat{A} \otimes \hat{B}$ is defined by

$$(\hat{A} \otimes \hat{B})(|\phi\rangle_A \otimes |\chi\rangle_B) = (\hat{A}|\phi\rangle_A) \otimes (\hat{B}|\chi\rangle_B), \quad (1.91)$$

and extended by linearity. In the product basis, its matrix elements read

$$\langle a_i, b_j | \hat{A} \otimes \hat{B} | a_{i'}, b_{j'} \rangle = \langle a_i | \hat{A} | a_{i'} \rangle \langle b_j | \hat{B} | b_{j'} \rangle. \quad (1.92)$$

Important special cases are the local operators $\hat{A} \otimes \mathbb{1}_B$ and $\mathbb{1}_A \otimes \hat{B}$, which act nontrivially only on one subsystem. For instance,

$$(\hat{A} \otimes \mathbb{1}_B)(|\phi\rangle_A \otimes |\chi\rangle_B) = (\hat{A}|\phi\rangle_A) \otimes |\chi\rangle_B. \quad (1.93)$$

If the global state is described by the density operator $\hat{\rho}_{AB}$, the expectation value of a local observable acting on A only is

$$\langle \hat{A} \otimes \mathbb{1}_B \rangle = \text{Tr}_{AB} [\hat{\rho}_{AB} (\hat{A} \otimes \mathbb{1}_B)]. \quad (1.94)$$

This motivates the introduction of reduced states and partial trace.

1.6.2 Local operators, reduced states, and partial trace

Let $\{|b_j\rangle\}$ be an orthonormal basis of $\mathcal{H}_B$. The partial trace over subsystem B is the map

$$\text{Tr}_B : \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B) \rightarrow \mathcal{B}(\mathcal{H}_A) \quad (1.95)$$

defined by

$$\text{Tr}_B(\hat{X}_{AB}) = \sum_j (\mathbb{1}_A \otimes \langle b_j|) \hat{X}_{AB} (\mathbb{1}_A \otimes |b_j\rangle). \quad (1.96)$$

The reduced states are then

$$\hat{\rho}_A = \text{Tr}_B(\hat{\rho}_{AB}), \quad \hat{\rho}_B = \text{Tr}_A(\hat{\rho}_{AB}). \quad (1.97)$$

It is important to verify that the reduced state reproduces all local expectation values. Writing

$$\hat{\rho}_{AB} = \sum_{i,i',j,j'} \rho_{ii';jj'} |a_i\rangle\langle a_{i'}| \otimes |b_j\rangle\langle b_{j'}|, \quad (1.98)$$

one finds

$$\text{Tr}_B(\hat{\rho}_{AB}) = \sum_{i,i',j} \rho_{ii';jj} |a_i\rangle\langle a_{i'}|. \quad (1.99)$$

Therefore,

$$\text{Tr}_A(\hat{\rho}_A \hat{A}) = \sum_{i,i',j} \rho_{ii';jj} \langle a_{i'} | \hat{A} | a_i \rangle, \quad (1.100)$$

while

$$\text{Tr}_{AB} \left[\hat{\rho}_{AB} (\hat{A} \otimes \mathbb{1}_B) \right] = \sum_{i,i',j,j'} \rho_{ii';jj'} \langle a_{i'} | \hat{A} | a_i \rangle \langle b_{j'} | b_j \rangle = \sum_{i,i',j} \rho_{ii';jj} \langle a_{i'} | \hat{A} | a_i \rangle. \quad (1.101)$$

Hence,

$$\text{Tr}_A(\hat{\rho}_A \hat{A}) = \text{Tr}_{AB} \left[\hat{\rho}_{AB} (\hat{A} \otimes \mathbb{1}_B) \right]. \quad (1.102)$$

Thus, the reduced density operator contains all information relevant to local measurements on subsystem A .

As a simple example, if the global state is a product state $\hat{\rho}_{AB} = \hat{\rho}_A \otimes \hat{\rho}_B$, then

$$\text{Tr}_B(\hat{\rho}_A \otimes \hat{\rho}_B) = \hat{\rho}_A \cdot \text{Tr}(\hat{\rho}_B) = \hat{\rho}_A, \quad (1.103)$$

since $\text{Tr}(\hat{\rho}_B) = 1$.

1.6.3 Bell basis and parity sectors

We now introduce the Bell basis, which plays a central role in two-qubit quantum information. It consists of the four maximally entangled states

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}} (|00\rangle \pm |11\rangle), \quad |\Psi^\pm\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle). \quad (1.104)$$

These states form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ and they explicit a simple parity structure. Defining the parity operator

$$\Pi_z = \sigma_z \otimes \sigma_z, \quad (1.105)$$

one finds

$$\Pi_z |\Phi^\pm\rangle = + |\Phi^\pm\rangle, \quad \Pi_z |\Psi^\pm\rangle = - |\Psi^\pm\rangle. \quad (1.106)$$

Hence, the two-qubit Hilbert space decomposes as

$$\mathcal{H}_2 = \mathcal{H}_{\text{even}} \oplus \mathcal{H}_{\text{odd}}, \quad (1.107)$$

with

$$\mathcal{H}_{\text{even}} = \text{span}\{|00\rangle, |11\rangle\}, \quad \mathcal{H}_{\text{odd}} = \text{span}\{|01\rangle, |10\rangle\}. \quad (1.108)$$

State	Computational form	Parity subspace	Eigenvalue of Π_z
$ \Phi^+\rangle$	$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	even	+1
$ \Phi^-\rangle$	$\frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$	even	+1
$ \Psi^+\rangle$	$\frac{1}{\sqrt{2}}(01\rangle + 10\rangle)$	odd	-1
$ \Psi^-\rangle$	$\frac{1}{\sqrt{2}}(01\rangle - 10\rangle)$	odd	-1

Table 1.1: Bell basis and parity sectors for two qubits.

The Bell basis is useful because it makes maximally entangled states explicit, diagonalizes relevant two-qubit observables, and naturally separates even and odd parity sectors. These notions will reappear later in the thesis in connection with parity measurements and monitored dynamics.

1.6.4 Separable and entangled states: definition and methods

We may now introduce the notion of separability. A pure bipartite state $|\Psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ is called separable if it can be written as $|\Psi\rangle_{AB} = |\phi\rangle_A \otimes |\chi\rangle_B$. If no such decomposition exists, the state is entangled. For mixed states, the definition is more general: a bipartite state $\hat{\rho}_{AB}$ is separable if it can be written as a convex combination of product states,

$$\hat{\rho}_{AB} = \sum_k p_k \hat{\rho}_A^{(k)} \otimes \hat{\rho}_B^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1. \quad (1.109)$$

If no decomposition of the form (1.109) exists, the state is entangled [14].

This distinction is conceptually fundamental. In a separable state, all correlations may be understood as arising from classical ignorance about local preparations. In an entangled state, by contrast, the global state contains correlations that cannot be reduced to any classical mixture of local states. Entanglement is therefore not simply a stronger form of correlation, but a genuinely nonclassical feature of the tensor-product structure of quantum theory. It is precisely this possibility that makes composite quantum systems radically different from their classical counterparts and turns entanglement into a central resource in quantum information, quantum metrology, and many-body physics.

A simple example of separable pure state is $|0\rangle \otimes |1\rangle = |01\rangle$. A standard example of entangled state, see (1.104)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle). \quad (1.110)$$

Indeed, if $|\Phi^+\rangle$ were a product state, one could write it as $(\alpha|0\rangle + \beta|1\rangle) \otimes (\gamma|0\rangle + \delta|1\rangle)$, which expands as

$$\alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle. \quad (1.111)$$

Comparison with Eq. (1.110) would require

$$\alpha\delta = 0, \quad \beta\gamma = 0, \quad \alpha\gamma = \beta\delta = \frac{1}{\sqrt{2}}, \quad (1.112)$$

which is impossible. Hence, $|\Phi^+\rangle$ is entangled.

For pure bipartite states, the structure of entanglement is completely characterized by the Schmidt decomposition. Let

$$|\Psi\rangle_{AB} = \sum_{i,j} c_{ij} |a_i\rangle \otimes |b_j\rangle. \quad (1.113)$$

By applying the singular value decomposition to the coefficient matrix $C = (c_{ij})$, one can always find orthonormal sets $\{|u_n\rangle\} \subset \mathcal{H}_A$ and $\{|v_n\rangle\} \subset \mathcal{H}_B$ such that

$$|\Psi\rangle_{AB} = \sum_{n=1}^r \sqrt{\lambda_n} |u_n\rangle \otimes |v_n\rangle, \quad (1.114)$$

where

$$\lambda_n \geq 0, \quad \sum_{n=1}^r \lambda_n = 1. \quad (1.115)$$

The integer r is called the Schmidt rank. The reduced density operators are then

$$\hat{\rho}_A = \text{Tr}_B(|\Psi\rangle\langle\Psi|_{AB}) = \sum_{n=1}^r \lambda_n |u_n\rangle\langle u_n|, \quad (1.116)$$

and similarly

$$\hat{\rho}_B = \sum_{n=1}^r \lambda_n |v_n\rangle\langle v_n|. \quad (1.117)$$

Hence, $\hat{\rho}_A$ and $\hat{\rho}_B$ have the same nonzero spectrum.

The Schmidt decomposition immediately implies that a pure bipartite state is separable if and only if its Schmidt rank is equal to one:

$$|\Psi\rangle_{AB} \text{ separable} \iff r = 1. \quad (1.118)$$

Equivalently, a pure bipartite state is entangled if and only if $r > 1$. The Schmidt form also makes clear why entanglement is directly connected with the mixedness of the reduced states. If the global state is pure and separable, then the reduced states are pure.

By contrast, if more than one Schmidt coefficient is nonzero, both reduced states are mixed. For instance, for the Bell state $|\Phi^+\rangle$ one has

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} |0\rangle \otimes |0\rangle + \frac{1}{\sqrt{2}} |1\rangle \otimes |1\rangle, \quad (1.119)$$

so that $\hat{\rho}_A = \hat{\rho}_B = \mathbb{1}/2$.

A natural quantity associated with a quantum state is the von Neumann entropy,

$$S(\hat{\rho}) = -\text{Tr}(\hat{\rho} \log \hat{\rho}). \quad (1.120)$$

If $\hat{\rho} = \sum_n \lambda_n |\varphi_n\rangle\langle\varphi_n|$, then

$$S(\hat{\rho}) = -\sum_n \lambda_n \log \lambda_n. \quad (1.121)$$

In particular, $S(\hat{\rho}) = 0$ if and only if $\hat{\rho}$ is pure. For a pure bipartite state $|\Psi\rangle\langle\Psi|_{AB}$, the reduced entropies satisfy

$$S(\hat{\rho}_A) = S(\hat{\rho}_B), \quad (1.122)$$

and this common value quantifies bipartite entanglement. In particular, separable pure states have zero reduced entropy, whereas Bell states have entropy $\log 2$.

For mixed states, deciding separability is in general much harder. One of the most important criteria is the partial-transposition criterion introduced by Peres [15]. Given a bipartite operator

$$\hat{\rho}_{AB} = \sum_{i,i',j,j'} \rho_{ii';jj'} |a_i\rangle\langle a_{i'}| \otimes |b_j\rangle\langle b_{j'}|, \quad (1.123)$$

its partial transpose with respect to subsystem B is defined as

$$\hat{\rho}_{AB}^{T_B} = \sum_{i,i',j,j'} \rho_{ii';jj'} |a_i\rangle\langle a_{i'}| \otimes |b_{j'}\rangle\langle b_j|. \quad (1.124)$$

If $\hat{\rho}_{AB}$ is separable and can be written as in Eq. (1.109), then

$$\hat{\rho}_{AB}^{T_B} = \sum_k p_k \hat{\rho}_A^{(k)} \otimes \left(\hat{\rho}_B^{(k)}\right)^T, \quad (1.125)$$

which is again positive, since transposition preserves positivity. Therefore,

$$\hat{\rho}_{AB} \text{ separable} \implies \hat{\rho}_{AB}^{T_B} \geq 0. \quad (1.126)$$

For 2×2 and 2×3 systems, the Horodecki family [14] proved that this condition is also sufficient:

$$\hat{\rho}_{AB}^{T_B} \geq 0 \iff \hat{\rho}_{AB} \text{ separable}, \quad d_A d_B \in \{4, 6\}. \quad (1.127)$$

In higher dimensions, however, PPT is only a necessary but not sufficient criterion.

A simple two-qubit example is again provided by the Bell state $|\Phi^+\rangle$. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$,

$$\hat{\rho}_{\Phi^+} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad (1.128)$$

while its partial transpose is

$$\hat{\rho}_{\Phi^+}^{T_B} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (1.129)$$

whose eigenvalues are $\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\}$. Since one eigenvalue is negative, the state is entangled.

A complementary and experimentally useful approach is provided by entanglement witnesses. A Hermitian operator $\hat{W}$ is called an entanglement witness if

$$\text{Tr}(\hat{W}\hat{\sigma}) \geq 0 \quad \text{for all separable states } \hat{\sigma}, \quad (1.130)$$

while there exists at least one entangled state $\hat{\rho}$ such that

$$\text{Tr}(\hat{W}\hat{\rho}) < 0. \quad (1.131)$$

Thus, a negative expectation value of $\hat{W}$ certifies entanglement. For the Bell state $|\Phi^+\rangle$, a simple witness is

$$\hat{W}_{\Phi^+} = \frac{1}{2} - |\Phi^+\rangle\langle\Phi^+|, \quad (1.132)$$

for which

$$\text{Tr}(\hat{W}_{\Phi^+} |\Phi^+\rangle\langle\Phi^+|) = -\frac{1}{2} < 0. \quad (1.133)$$

1.6.5 Concurrence for two-qubit states

For two-qubit systems, one of the most widely used entanglement measures is the concurrence. This quantity is especially important because, unlike the general mixed-state separability problem, it admits a closed analytical expression for arbitrary two-qubit states [16, 17]. It is therefore particularly useful whenever one needs an explicit and computable measure of entanglement, both for pure states and for generic mixed states. We consider a two-qubit system with Hilbert space $\mathcal{H}_{AB} = \mathbb{C}^2 \otimes \mathbb{C}^2$ and computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Let

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle, \quad |a|^2 + |b|^2 + |c|^2 + |d|^2 = 1 \quad (1.134)$$

be a pure state. The spin-flipped state is defined as

$$|\tilde{\psi}\rangle = (\sigma_y \otimes \sigma_y) |\psi^*\rangle, \quad (1.135)$$

where the complex conjugation is taken in the computational basis. The concurrence is then

$$C(|\psi\rangle) = \left| \langle \psi | \tilde{\psi} \rangle \right| = 2|ad - bc|. \quad (1.136)$$

This expression already shows the basic behavior of the measure. One has

$$0 \leq C(|\psi\rangle) \leq 1, \quad (1.137)$$

with

$$C(|\psi\rangle) = 0 \iff |\psi\rangle \text{ is separable}, \quad C(|\psi\rangle) = 1 \iff |\psi\rangle \text{ is maximally entangled.} \quad (1.138)$$

In particular, $C(|\Phi^+\rangle) = 1$, whereas every product state has vanishing concurrence because $ad - bc = 0$.

The quantity $ad - bc$ is the determinant of the 2×2 coefficient matrix associated with the pure state. Its vanishing is therefore equivalent to the factorization of the state vector, which makes the relation between concurrence and separability particularly transparent in the pure-state case. The spin-flip operation entering Eq. (1.135) is also natural from the two-qubit point of view, since it maps each local basis vector to its orthogonal counterpart up to a phase and allows one to construct a scalar quantity sensitive to entanglement.

The concurrence of a pure two-qubit state can also be expressed in terms of the reduced state:

$$C(|\psi\rangle) = 2\sqrt{\det \hat{\rho}_A} = \sqrt{2(1 - \text{Tr}(\hat{\rho}_A^2))}, \quad \hat{\rho}_A = \text{Tr}_B(|\psi\rangle\langle\psi|). \quad (1.139)$$

Thus, for pure bipartite states, concurrence is directly related to the mixedness of the reduced state. This is fully consistent with the Schmidt-decomposition picture discussed above: a pure bipartite state is entangled precisely when its reduced states are mixed. For mixed states, the situation is conceptually more subtle. A mixed state may admit many different pure-state decompositions, and an entanglement measure should not depend on a particular ensemble representation. For this reason, concurrence is defined through the convex-roof construction [16]:

$$C(\hat{\rho}) = \min_{\{p_k, |\psi_k\rangle\}} \sum_k p_k C(|\psi_k\rangle), \quad (1.140)$$

where the minimum is taken over all decompositions

$$\hat{\rho} = \sum_k p_k |\psi_k\rangle\langle\psi_k|. \quad (1.141)$$

This definition is natural but, in general, difficult to evaluate explicitly. The remarkable result proved by Wootters is that for two qubits the minimization can be solved in closed form [16].

Defining the spin-flipped density operator

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y) \hat{\rho}^* (\sigma_y \otimes \sigma_y), \quad (1.142)$$

and the operator

$$\hat{R} = \sqrt{\sqrt{\hat{\rho}} \tilde{\rho} \sqrt{\hat{\rho}}}, \quad (1.143)$$

let $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0$ be the eigenvalues of $\hat{R}$. Then

$$C(\hat{\rho}) = \max \{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}. \quad (1.144)$$

Equivalently, if μ_i are the eigenvalues of $\hat{\rho}\tilde{\rho}$, one may take $\lambda_i = \sqrt{\mu_i}$ ordered in decreasing way. Thus, given a generic two-qubit density matrix, the calculation proceeds through the chain

$$\hat{\rho} \longrightarrow \tilde{\rho} \longrightarrow \hat{\rho}\tilde{\rho} \longrightarrow \{\mu_i\} \longrightarrow \{\lambda_i\} \longrightarrow C(\hat{\rho}). \quad (1.145)$$

This provides a complete analytical measure of entanglement for arbitrary two-qubit states.

Several properties are worth emphasizing. First,

$$0 \leq C(\hat{\rho}) \leq 1. \quad (1.146)$$

Second, in the two-qubit case,

$$C(\hat{\rho}) = 0 \iff \hat{\rho} \text{ is separable.} \quad (1.147)$$

Thus, concurrence is not only an entanglement monotone, but also a complete separability criterion for two qubits. Third, concurrence is invariant under local unitary transformations,

$$C\left[(\hat{U}_A \otimes \hat{U}_B)\hat{\rho}(\hat{U}_A^\dagger \otimes \hat{U}_B^\dagger)\right] = C(\hat{\rho}), \quad (1.148)$$

as expected for any physically meaningful measure of bipartite entanglement.

Finally, concurrence is directly related to the entanglement of formation. For two qubits, Wootters showed that

$$E_F(\hat{\rho}) = h\left(\frac{1 + \sqrt{1 - C(\hat{\rho})^2}}{2}\right), \quad (1.149)$$

where

$$h(x) = -x \log_2 x - (1 - x) \log_2 (1 - x) \quad (1.150)$$

is the binary entropy. This relation is particularly important because it shows that concurrence is not just an auxiliary quantity: it provides a direct route to one of the most physically meaningful bipartite entanglement measures.

The concurrence is therefore especially convenient in the two-qubit setting. It vanishes exactly on separable states, reaches its maximal value on Bell states, admits a simple expression for pure states, and, remarkably, remains analytically computable for arbitrary mixed states. For this reason, it will be one of the main tools employed throughout this thesis whenever two-qubit entanglement needs to be quantified explicitly.

1.6.6 Generic multipartite systems and collective spin observables

The generalization from bipartite to generic multipartite systems is conceptually straightforward at the level of Hilbert spaces, but it immediately leads to a much richer mathematical and physical structure. For N distinguishable subsystems with Hilbert spaces $\mathcal{H}_1, \dots, \mathcal{H}_N$, the global Hilbert space is the tensor product

$$\mathcal{H}^{(N)} = \mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_N. \quad (1.151)$$

In the case of N qubits, each local Hilbert space is $\mathbb{C}^2$, and therefore

$$\mathcal{H}^{(N)} = (\mathbb{C}^2)^{\otimes N}, \quad \dim \mathcal{H}^{(N)} = 2^N. \quad (1.152)$$

A natural orthonormal basis is provided by the computational product basis

$$|x_1, \dots, x_N\rangle = |x_1\rangle \otimes \cdots \otimes |x_N\rangle, \quad x_k \in \{0, 1\}. \quad (1.153)$$

These vectors are orthonormal, since

$$\langle x_1, \dots, x_N | y_1, \dots, y_N \rangle = \prod_{k=1}^N \langle x_k | y_k \rangle = \prod_{k=1}^N \delta_{x_k y_k}. \quad (1.154)$$

Accordingly, any pure state of the N -qubit system may be expanded as

$$|\Psi\rangle = \sum_{x_1, \dots, x_N \in \{0, 1\}} c_{x_1 \dots x_N} |x_1, \dots, x_N\rangle, \quad (1.155)$$

with normalization condition

$$\sum_{x_1, \dots, x_N} |c_{x_1 \dots x_N}|^2 = 1. \quad (1.156)$$

At the multipartite level, local and nonlocal operators must be carefully distinguished.

If $\hat{A}$ is an operator acting on the Hilbert space of a single subsystem, say the k -th one, then the corresponding local operator on the full Hilbert space is defined as

$$\hat{A}^{(k)} = \mathbb{1}^{\otimes(k-1)} \otimes \hat{A} \otimes \mathbb{1}^{\otimes(N-k)}. \quad (1.157)$$

This means that $\hat{A}^{(k)}$ acts as $\hat{A}$ on the k -th factor and as the identity on all the others. Its action on a computational-basis vector is therefore

$$\hat{A}^{(k)} |x_1, \dots, x_k, \dots, x_N\rangle = |x_1\rangle \otimes \cdots \otimes \hat{A} |x_k\rangle \otimes \cdots \otimes |x_N\rangle. \quad (1.158)$$

For example, let $N = 3$ and consider the operator $\sigma_x^{(2)} = \mathbb{1} \otimes \sigma_x \otimes \mathbb{1}$. Since $\sigma_x |0\rangle = |1\rangle$ and $\sigma_x |1\rangle = |0\rangle$, one has

$$\sigma_x^{(2)} |0, 0, 1\rangle = |0, 1, 1\rangle, \quad \sigma_x^{(2)} |1, 1, 0\rangle = |1, 0, 0\rangle. \quad (1.159)$$

Similarly, using the convention $\sigma_z |0\rangle = |0\rangle$ and $\sigma_z |1\rangle = -|1\rangle$, one finds

$$\sigma_z^{(3)} |0, 1, 0\rangle = |0, 1, 0\rangle, \quad \sigma_z^{(3)} |0, 1, 1\rangle = -|0, 1, 1\rangle. \quad (1.160)$$

Thus, local operators act only on the corresponding slot of the tensor product, leaving the remaining factors unchanged.

Their action on a generic state follows by linearity. For instance, if

$$|\Psi\rangle = a |000\rangle + b |011\rangle + c |101\rangle + d |110\rangle, \quad (1.161)$$

then

$$\sigma_x^{(2)} |\Psi\rangle = a |010\rangle + b |001\rangle + c |111\rangle + d |100\rangle, \quad (1.162)$$

while

$$\sigma_z^{(1)} |\Psi\rangle = a |000\rangle + b |011\rangle - c |101\rangle - d |110\rangle. \quad (1.163)$$

These explicit examples are useful because they show that a local operator on a multipartite system is not a small operator: it still acts on the full Hilbert space, but in a factorized way.

More generally, if $\hat{A}$ acts on subsystem k and $\hat{B}$ acts on subsystem ℓ , with $k \neq \ell$, then one defines the corresponding two-body operator as

$$\hat{A}^{(k)} \hat{B}^{(\ell)} = \mathbb{1}^{\otimes(k-1)} \otimes \hat{A} \otimes \mathbb{1}^{\otimes(\ell-k-1)} \otimes \hat{B} \otimes \mathbb{1}^{\otimes(N-\ell)}, \quad (1.164)$$

assuming $k < \ell$. Such operators are central in interacting many-body models, correlation functions, and spin Hamiltonians. For example, for $N = 3$,

$$\sigma_z^{(1)} \sigma_z^{(3)} |x_1, x_2, x_3\rangle = \lambda_{x_1} \lambda_{x_3} |x_1, x_2, x_3\rangle, \quad (1.165)$$

where $\lambda_0 = +1$ and $\lambda_1 = -1$. Thus, $\sigma_z^{(1)} \sigma_z^{(3)}$ measures the relative parity of the first and third qubits.

Reduced states are also naturally generalized. If $I \subseteq \{1, \dots, N\}$ denotes a subset of subsystems, one may define the reduced state of that subset by tracing out the complementary degrees of freedom:

$$\hat{\rho}_I = \text{Tr}_{\bar{I}}(\hat{\rho}). \quad (1.166)$$

In this way, local properties, marginal statistics, and subsystem correlations may be studied inside the full multipartite state. As in the bipartite case, the reduced state contains all information relevant to observables acting only on the chosen subset.

The notion of separability becomes substantially richer in the multipartite setting. A pure state is said to be fully separable if it can be written as

$$|\Psi\rangle = |\phi_1\rangle \otimes \dots \otimes |\phi_N\rangle. \quad (1.167)$$

Mixed states are fully separable if they are convex combinations of such product states. However, one may also encounter states that are separable with respect to some bipartitions but not with respect to others. This leads to several inequivalent notions of multipartite entanglement. In particular, a state is said to possess genuine multipartite entanglement if it cannot be written as a convex combination of states that are separable across some bipartition [18, 19]. Thus, the multipartite case is not merely a larger version of the bipartite one: it exhibits a genuinely more intricate hierarchy of correlation structures.

A simple example illustrates this point. For three qubits, the state

$$|\Psi\rangle = |0\rangle \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}} \quad (1.168)$$

is entangled, but not genuinely tripartite entangled, since the first qubit is completely factorized from the remaining pair. By contrast, the GHZ state

$$|\text{GHZ}_3\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}} \quad (1.169)$$

cannot be factorized with respect to any bipartition and is therefore genuinely tripartite entangled.

For the purposes of the present thesis, a particularly important family of observables is given by the collective spin operators

$$\hat{J}_\alpha = \frac{1}{2} \sum_{k=1}^N \sigma_\alpha^{(k)}, \quad \alpha \in \{x, y, z\}. \quad (1.170)$$

These are sums of local operators, and therefore represent global observables obtained by coherently combining the contributions of all subsystems. Their action on a basis vector is determined term by term. For example,

$$\hat{J}_z |x_1, \dots, x_N\rangle = \frac{1}{2} \sum_{k=1}^N \lambda_{x_k} |x_1, \dots, x_N\rangle, \quad (1.171)$$

with $\lambda_0 = +1$ and $\lambda_1 = -1$. Thus, each computational-basis vector is an eigenvector of $\hat{J}_z$, with eigenvalue equal to one half of the difference between the number of qubits in $|0\rangle$ and the number of qubits in $|1\rangle$.

The operators $\hat{J}_x, \hat{J}_y, \hat{J}_z$ satisfy the angular-momentum commutation relations

$$[\hat{J}_x, \hat{J}_y] = i\hat{J}_z, \quad [\hat{J}_y, \hat{J}_z] = i\hat{J}_x, \quad [\hat{J}_z, \hat{J}_x] = i\hat{J}_y, \quad (1.172)$$

and therefore provide the natural language for the description of collective spin observables, permutation-symmetric states, and spin-squeezing properties in many-qubit systems.

It is worth stressing that collective operators are global, but not generic. They are built from sums of identical local operators and therefore respect the permutation symmetry of the subsystems. This will be especially important later, since many states of interest in quantum metrology and collective open-system dynamics naturally live in the symmetric sector of the full Hilbert space.

Finally, one may also construct collective two-body quantities, such as

$$\hat{J}_\alpha^2 = \frac{1}{4} \sum_{k,\ell=1}^N \sigma_\alpha^{(k)} \sigma_\alpha^{(\ell)} = \frac{N}{4} \mathbb{1} + \frac{1}{4} \sum_{\substack{k,\ell=1 \\ k \neq \ell}}^N \sigma_\alpha^{(k)} \sigma_\alpha^{(\ell)}, \quad (1.173)$$

where the identity contribution comes from $(\sigma_\alpha^{(k)})^2 = \mathbb{1}$. This expansion makes explicit the connection between collective observables and two-body correlations. It is precisely this link that will later make collective variances useful probes of multipartite entanglement and spin squeezing.

1.7 Spin squeezing

Spin squeezing is one of the most important collective manifestations of nonclassical correlations in many-spin systems. It was originally introduced by Kitagawa and Ueda as the spin analogue of quadrature squeezing in bosonic systems [20], and was later reinterpreted by Wineland *et al.* as a directly metrological resource for phase estimation

beyond the standard quantum limit [21]. The modern viewpoint, clearly reviewed in [22], is that spin squeezing lies at the intersection of three closely related notions: redistribution of collective quantum fluctuations, multipartite entanglement, and quantum-enhanced metrology.

1.7.1 Collective spin operators and coherent spin states

For a system of N spin-1/2 particles, the collective spin operators are

$$\hat{J}_\alpha = \frac{1}{2} \sum_{k=1}^N \sigma_\alpha^{(k)}, \quad \alpha \in \{x, y, z\}, \quad (1.174)$$

and satisfy the usual angular-momentum commutation relations. For any unit vector $\mathbf{n}$, we write $\hat{J}_\mathbf{n} = \mathbf{n} \cdot \hat{\mathbf{J}}$.

The natural reference family is given by coherent spin states. In the present convention, a generic coherent spin state reads

$$|\theta, \phi\rangle = \bigotimes_{k=1}^N \left(e^{i\phi} \sin \frac{\theta}{2} |1\rangle_k + \cos \frac{\theta}{2} |0\rangle_k \right), \quad (1.175)$$

with mean-spin direction

$$\mathbf{n}_0 = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \quad (1.176)$$

For these states one has

$$\langle \hat{\mathbf{J}} \rangle = \frac{N}{2} \mathbf{n}_0, \quad (1.177)$$

so that the mean-spin length is maximal, $|\langle \hat{\mathbf{J}} \rangle| = N/2$. Moreover, for every direction $\mathbf{n}_\perp$ orthogonal to $\mathbf{n}_0$,

$$(\Delta \hat{J}_{\mathbf{n}_\perp})^2 = \frac{N}{4}. \quad (1.178)$$

Thus, coherent spin states define the standard quantum limit for collective-spin fluctuations.

1.7.2 Kitagawa-Ueda and Wineland definitions

Assuming $|\langle \hat{\mathbf{J}} \rangle| \neq 0$, we define the mean-spin direction as

$$\mathbf{n}_0 = \frac{\langle \hat{\mathbf{J}} \rangle}{|\langle \hat{\mathbf{J}} \rangle|}. \quad (1.179)$$

Spin squeezing is then characterized through the minimal variance in the plane orthogonal to $\mathbf{n}_0$:

$$(\Delta \hat{J}_\perp)^2_{\min} = \min_{\mathbf{n}_\perp \perp \mathbf{n}_0} (\Delta \hat{J}_{\mathbf{n}_\perp})^2. \quad (1.180)$$

If the mean spin points along the z axis, one finds

$$(\Delta \hat{J}_\perp)^2_{\min} = \frac{1}{2} \left[\langle \hat{J}_x^2 + \hat{J}_y^2 \rangle - \sqrt{\langle \hat{J}_x^2 - \hat{J}_y^2 \rangle^2 + \langle [\hat{J}_x, \hat{J}_y]_+ \rangle^2} \right]. \quad (1.181)$$

The Kitagawa-Ueda parameter is defined as [20, 22]

$$\xi_{\text{KU}}^2 = \frac{4}{N}(\Delta\hat{J}_\perp)_{\min}^2. \quad (1.182)$$

A state is spin squeezed in the Kitagawa-Ueda sense if $\xi_{\text{KU}}^2 < 1$. This criterion compares the minimal transverse variance with the coherent-spin reference value $N/4$, and therefore captures the purely fluctuation-based aspect of squeezing.

The Wineland parameter is instead motivated by Ramsey interferometry [21, 22]. For optimal small-angle phase estimation one obtains

$$\Delta\phi = \frac{(\Delta\hat{J}_\perp)_{\min}}{|\langle\hat{\mathbf{J}}\rangle|}, \quad (1.183)$$

which, compared with the shot-noise limit $\Delta\phi_{\text{SQL}} = 1/\sqrt{N}$, leads to

$$\xi_{\text{W}}^2 = \frac{N(\Delta\hat{J}_\perp)_{\min}^2}{|\langle\hat{\mathbf{J}}\rangle|^2}. \quad (1.184)$$

The condition $\xi_{\text{W}}^2 < 1$ indicates metrological gain beyond the standard quantum limit. The two parameters are related by

$$\xi_{\text{W}}^2 = \frac{N^2}{4|\langle\hat{\mathbf{J}}\rangle|^2} \xi_{\text{KU}}^2, \quad (1.185)$$

so that $\xi_{\text{KU}}^2 \leq \xi_{\text{W}}^2$, since $|\langle\hat{\mathbf{J}}\rangle| \leq N/2$. Thus, Wineland squeezing is the stronger requirement.

For a coherent spin state, one has $(\Delta\hat{J}_\perp)_{\min}^2 = N/4$ and $|\langle\hat{\mathbf{J}}\rangle| = N/2$, hence

$$\xi_{\text{KU}}^2 = \xi_{\text{W}}^2 = 1. \quad (1.186)$$

Thus, coherent spin states are unsqueezed and define the natural classical benchmark.

An instructive family is given by Dicke states $|J, m\rangle$ with $J = N/2$. For these states, $\langle\hat{J}_z\rangle = m$ and $\langle\hat{J}_x\rangle = \langle\hat{J}_y\rangle = 0$. For $m \neq 0$, the mean spin points along z , and one finds

$$(\Delta\hat{J}_x)^2 = (\Delta\hat{J}_y)^2 = \frac{1}{2} [J(J+1) - m^2], \quad (1.187)$$

so that

$$\xi_{\text{KU}}^2 = \frac{2}{N} [J(J+1) - m^2]. \quad (1.188)$$

For $m = \pm J$, one recovers $\xi_{\text{KU}}^2 = 1$, consistently with the fact that $|J, \pm J\rangle$ are coherent spin states. For $|m| < J$, one obtains $\xi_{\text{KU}}^2 > 1$. In particular, Dicke states show that entanglement and spin squeezing are not equivalent notions.

A further caveat is that the original mean-spin-based construction (Wineland case!) becomes problematic when $|\langle\hat{\mathbf{J}}\rangle| = 0$. In that case the direction $\mathbf{n}_0$ is not defined, and the interpretation of both ξ_{KU}^2 and ξ_{W}^2 must be handled with care. This is one of the reasons why more general squeezing and entanglement criteria were later developed [22, 23].

Lastly, their original work, Kitagawa and Ueda also introduced two paradigmatic nonlinear Hamiltonians capable of generating spin squeezing from an initial coherent spin state [20]:

$$\hat{H}_{\text{OAT}} = \chi \hat{J}_z^2, \quad \hat{H}_{\text{TAT}} = \chi(\hat{J}_x^2 - \hat{J}_y^2), \quad (1.189)$$

up to trivial relabelings of the axes. The nonlinear dynamics shears the initially isotropic fluctuation profile in the transverse plane, producing a squeezed axis and an anti-squeezed one. The one-axis twisting model is analytically tractable and experimentally widespread, while the two-axis counter-twisting model is more efficient and can approach Heisenberg scaling more closely, although it is harder to engineer.

1.7.3 Spin squeezing and symmetric states

We now discuss an important result due to Wang and Sanders [24], which connects the Kitagawa-Ueda squeezing parameter to negative pairwise correlations and pairwise entanglement for symmetric multiqubit states.

Let $\mathbf{n}_\perp$ be a direction orthogonal to the mean spin, and define the local Pauli operator

$$\sigma_{\mathbf{n}_\perp}^{(i)} = \mathbf{n}_\perp \cdot \vec{\sigma}^{(i)}. \quad (1.190)$$

For a symmetric state, all one-body expectations $\langle \sigma_{\mathbf{n}_\perp}^{(i)} \rangle$ are equal, and all two-body correlations $\langle \sigma_{\mathbf{n}_\perp}^{(i)} \otimes \sigma_{\mathbf{n}_\perp}^{(j)} \rangle$ with $i \neq j$ are identical.

Lemma 1: For a symmetric separable state of N qubits, the pairwise transverse correlation is nonnegative:

$$\langle \sigma_{\mathbf{n}_\perp}^{(i)} \otimes \sigma_{\mathbf{n}_\perp}^{(j)} \rangle \geq 0, \quad i \neq j. \quad (1.191)$$

Proof. A symmetric separable state may be written as

$$\hat{\rho}_{\text{sep}} = \sum_k p_k \hat{\rho}^{(k)} \otimes \hat{\rho}^{(k)} \otimes \cdots \otimes \hat{\rho}^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1. \quad (1.192)$$

Its two-qubit reduced state is therefore

$$\hat{\rho}_{ij} = \sum_k p_k \hat{\rho}^{(k)} \otimes \hat{\rho}^{(k)}. \quad (1.193)$$

Hence,

$$\langle \sigma_{\mathbf{n}_\perp}^{(i)} \otimes \sigma_{\mathbf{n}_\perp}^{(j)} \rangle = \sum_k p_k \text{Tr}[(\hat{\rho}^{(k)} \otimes \hat{\rho}^{(k)})(\sigma_{\mathbf{n}_\perp} \otimes \sigma_{\mathbf{n}_\perp})]. \quad (1.194)$$

Since the tensor factors are identical, this becomes

$$\langle \sigma_{\mathbf{n}_\perp}^{(i)} \otimes \sigma_{\mathbf{n}_\perp}^{(j)} \rangle = \sum_k p_k (\text{Tr}[\hat{\rho}^{(k)} \sigma_{\mathbf{n}_\perp}])^2 \geq 0. \quad (1.195)$$

This proves the claim.

Proposition: For an arbitrary symmetric multiqubit state, the condition $\xi_{\text{KU}}^2 < 1$ implies pairwise entanglement.

Proof. For a direction $\mathbf{n}_\perp$ orthogonal to the mean spin, one has $\langle \hat{J}_{\mathbf{n}_\perp} \rangle = 0$, and therefore

$$(\Delta \hat{J}_{\mathbf{n}_\perp})^2 = \langle \hat{J}_{\mathbf{n}_\perp}^2 \rangle. \quad (1.196)$$

Now,

$$\hat{J}_{\mathbf{n}_\perp} = \frac{1}{2} \sum_{k=1}^N \sigma_{\mathbf{n}_\perp}^{(k)}, \quad (1.197)$$

hence

$$\hat{J}_{\mathbf{n}_\perp}^2 = \frac{1}{4} \sum_{k=1}^N (\sigma_{\mathbf{n}_\perp}^{(k)})^2 + \frac{1}{4} \sum_{\substack{k,\ell=1 \\ k \neq \ell}}^N \sigma_{\mathbf{n}_\perp}^{(k)} \sigma_{\mathbf{n}_\perp}^{(\ell)}. \quad (1.198)$$

Since $(\sigma_{\mathbf{n}_\perp}^{(k)})^2 = \mathbb{1}$, taking the expectation value yields

$$\langle \hat{J}_{\mathbf{n}_\perp}^2 \rangle = \frac{N}{4} + \frac{1}{4} \sum_{\substack{k,\ell=1 \\ k \neq \ell}}^N \langle \sigma_{\mathbf{n}_\perp}^{(k)} \sigma_{\mathbf{n}_\perp}^{(\ell)} \rangle. \quad (1.199)$$

By exchange symmetry, all pair correlations are equal, so for any $i \neq j$,

$$\langle \hat{J}_{\mathbf{n}_\perp}^2 \rangle = \frac{N}{4} + \frac{N(N-1)}{4} \langle \sigma_{\mathbf{n}_\perp}^{(i)} \otimes \sigma_{\mathbf{n}_\perp}^{(j)} \rangle. \quad (1.200)$$

Substituting this into the Kitagawa-Ueda definition gives

$$\xi_{\text{KU}}^2 = \frac{4}{N} (\Delta \hat{J}_{\mathbf{n}_\perp})^2 = 1 + (N-1) \langle \sigma_{\mathbf{n}_\perp}^{(i)} \otimes \sigma_{\mathbf{n}_\perp}^{(j)} \rangle. \quad (1.201)$$

Therefore,

$$\xi_{\text{KU}}^2 < 1 \iff \langle \sigma_{\mathbf{n}_\perp}^{(i)} \otimes \sigma_{\mathbf{n}_\perp}^{(j)} \rangle < 0. \quad (1.202)$$

By Lemma 1, such a negative pairwise correlation is impossible for a symmetric separable two-qubit reduction. Hence, if $\xi_{\text{KU}}^2 < 1$, the reduced two-qubit state must be entangled. This proves pairwise entanglement.

This result is particularly important because it shows that, for symmetric states, Kitagawa-Ueda spin squeezing is not merely a fluctuation-based property: it is already a direct witness of nonclassical two-body correlations. At the same time, the implication is one-way in general. There exist symmetric entangled states, such as Dicke states, that are not spin squeezed according to the standard criterion.

2

Open Quantum Systems and continuously monitored systems

The formulation presented in Chapter 1 provides the standard description of closed and isolated quantum systems. In that setting, the state is associated with a vector in Hilbert space, the dynamics is unitary, and measurements are described through the Born rule and the projection postulate. This framework is conceptually fundamental and mathematically consistent. At the same time, it relies on an idealization that is rarely realized in practice. Any realistic quantum system is, at least to some extent, coupled to additional degrees of freedom, whether they correspond to uncontrolled modes of the electromagnetic field, external reservoirs, measurement devices, or simply to the surrounding laboratory conditions [25, 26].

This observation is not merely of practical relevance. It marks a genuine conceptual extension of the standard theory. If the system under investigation interacts with external degrees of freedom, then its evolution is no longer described, in general, by a unitary operator acting only on its Hilbert space. The reduced dynamics may display decoherence, dissipation, irreversibility, and noise. These effects emerge naturally once one recognizes that the system of interest should be treated as a subsystem of a larger closed composite system.

For this reason, the theory of open quantum systems may be viewed as the natural continuation of the standard formulation of quantum mechanics. One distinguishes between the subsystem of interest, usually denoted by S , and an external environment E where the total system $S + E$ is still described within the ordinary postulates of quantum mechanics. However, the subsystem S is no longer isolated, and its state must be described through a reduced statistical operator.

2.1 Open quantum systems: formalism

To make this point more explicit, let us consider a quantum system S coupled to an environment E . The Hilbert space of the composite is

$$\mathcal{H}_{SE} = \mathcal{H}_S \otimes \mathcal{H}_E. \quad (2.1)$$

The total system $S + E$ is still assumed to be closed. Therefore, its dynamics is governed by a unitary evolution on $\mathcal{H}_{SE}$.

The total Hamiltonian is written as

$$\hat{H}_{SE} = \hat{H}_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \hat{H}_E + \hat{H}_{\text{int}}, \quad (2.2)$$

so the corresponding evolution of the global density operator is

$$\hat{\rho}_{SE}(t) = \hat{U}_{SE}(t, t_0) \hat{\rho}_{SE}(t_0) \hat{U}_{SE}^\dagger(t, t_0), \quad (2.3)$$

where $\hat{U}_{SE}(t, t_0)$ is unitary. Since the environment is not directly observed, the physically relevant description is obtained by tracing out its degrees of freedom:

$$\hat{\rho}_S(t) = \text{Tr}_E[\hat{\rho}_{SE}(t)]. \quad (2.4)$$

By combining Eqs. (2.3) and (2.4), one obtains

$$\hat{\rho}_S(t) = \text{Tr}_E \left[\hat{U}_{SE}(t, t_0) \hat{\rho}_{SE}(t_0) \hat{U}_{SE}^\dagger(t, t_0) \right]. \quad (2.5)$$

The total evolution is unitary, but not the reduced evolution of the subsystem. In the standard formulation for isolated systems, the evolution is described by a unitary operator acting on the Hilbert space of the system alone: such an evolution preserves purity, preserves the spectrum of the density operator, and is reversible. In particular, if the initial state is pure, it remains pure at all times.

By contrast, the reduced evolution in Eq. (2.5) is totally different, and it loses all these "good" properties. The reason is that the interaction term $\hat{H}_{\text{int}}$ creates correlations between S and E . Even if the total state remains pure, the subsystem may become mixed because part of the information about S is now stored in the correlations with the environment. In this sense, the non-unitary character of the reduced dynamics comes from the fact that one is describing only a part of a larger quantum system.

A first important consequence is that an initially pure state of the subsystem may evolve into a mixed state. This is impossible under unitary dynamics on $\mathcal{H}_S$, but it is completely natural for an open system. A second consequence is that the reduced evolution need not be reversible. The partial trace is not an invertible operation: once the environmental degrees of freedom are discarded, the information encoded in system-environment correlations is no longer accessible from the subsystem alone. A third consequence is that quantities such as purity (1.43) and von Neumann entropy (1.120) are no longer conserved for the reduced state, even though they are conserved for the total isolated system.

It is therefore natural to replace the unitary description on $\mathcal{H}_S$ with a more general dynamical law. In the simplest and most relevant situation, one assumes that the initial global state is factorized,

$$\hat{\rho}_{SE}(t_0) = \hat{\rho}_S(t_0) \otimes \hat{\rho}_E(t_0). \quad (2.6)$$

Under this assumption, the reduced evolution may be written as a map

$$\hat{\rho}_S(t) = \Phi_{t,t_0}[\hat{\rho}_S(t_0)], \quad (2.7)$$

where Φ_{t,t_0} is, in general, not unitary. Rather, it is a quantum dynamical map acting on density operators [25, 26].

This observation already shows why the standard postulates of closed-system quantum mechanics are not sufficient in the present context. The appropriate framework is instead based on generalized measurements and CPTP maps. Within this more general formulation, continuously monitored systems will arise as a further refinement, in which the open-system evolution is conditioned on the information extracted from the environment through a continuous measurement record.

2.1.1 Generalized measurements, the POVMs

The projective description of quantum measurements is sufficient when one deals with ideal measurements directly performed on the system Hilbert space. In many relevant situations, however, the measurement process is more involved. The system first interacts with additional degrees of freedom, such as an ancilla, a probe, or a measuring device, and only part of the total setup is eventually observed. In this case, the effective measurement acting on the system alone is no longer described, in general, by a family of orthogonal projectors [26, 25].

This is the natural setting in which generalized measurements arise. Their role is to describe the statistics of outcomes when the measurement is indirect, noisy, or only partially resolving. From this point of view, POVMs are not a modification introduced by hand. They are the effective description of standard measurements performed on a larger Hilbert space and then reduced to the subsystem of interest.

A more general framework is therefore provided by positive operator-valued measures, POVMs. A POVM is a set of positive operators $\{\hat{\Pi}_x\}$ satisfying

$$\hat{\Pi}_x \geq 0, \quad \sum_x \hat{\Pi}_x = \mathbb{1}. \quad (2.8)$$

The probability of obtaining the outcome x when the system is in the state $\hat{\rho}$ is

$$p_x = \text{Tr}[\hat{\rho} \hat{\Pi}_x]. \quad (2.9)$$

Projective measurements are recovered as a special case by taking $\Pi_x = \hat{P}_x$ where these are orthogonal projectors.

To describe the post-measurement state one introduces a set of operators $\{\hat{M}_x\}$, usually called detection operators or measurement operators, such that

$$\hat{\Pi}_x = \hat{M}_x^\dagger \hat{M}_x, \quad \sum_x \hat{M}_x^\dagger \hat{M}_x = \mathbb{1}. \quad (2.10)$$

In terms of these operators, the generalized Born rule may be written as

$$p_x = \text{Tr}[\hat{M}_x \hat{\rho} \hat{M}_x^\dagger] = \text{Tr}[\hat{\rho} \hat{M}_x^\dagger \hat{M}_x] = \text{Tr}[\hat{\rho} \hat{\Pi}_x]. \quad (2.11)$$

If the outcome x is recorded, the corresponding conditional state is

$$\hat{\rho}_x = \frac{\hat{M}_x \hat{\rho} \hat{M}_x^\dagger}{p_x} = \frac{\hat{M}_x \hat{\rho} \hat{M}_x^\dagger}{\text{Tr}[\hat{\rho} \hat{\Pi}_x]}. \quad (2.12)$$

If the measurement is performed but the outcome is not read out, the post-measurement state is

$$\tilde{\hat{\rho}} = \sum_x p_x \hat{\rho}_x = \sum_x \hat{M}_x \hat{\rho} \hat{M}_x^\dagger. \quad (2.13)$$

This formulation contains the projective one, but is strictly more general. In particular, orthogonality of the POVM elements is not required, and the number of outcomes is not bounded by the dimension of the Hilbert space. Moreover, for a fixed POVM the detection operators are not unique. Since each $\hat{\Pi}_x$ is positive, its square root is well defined, and one may always write

$$\hat{M}_x = \hat{U}_x \sqrt{\hat{\Pi}_x}, \quad (2.14)$$

where $\hat{U}_x$ is unitary on the support of $\hat{\Pi}_x$. This is simply the polar decomposition of $\hat{M}_x$. It shows that the POVM elements determine the probabilities, whereas the unitary part of the detection operators determines how the state is transformed after the measurement.

The measurement statistics is described by the POVM

$$\hat{\Pi}_n = |n\rangle \langle n|. \quad (2.15)$$

If the detector is demolitive, the mode is left in the vacuum after the detection, independently of the outcome, that is $\rho_n = |0\rangle\langle 0|$. The detection operators are thus described by:

$$\hat{M}_n = |0\rangle \langle n|. \quad (2.16)$$

indeed,

$$\hat{M}_n^\dagger \hat{M}_n = |n\rangle \langle 0|0\rangle \langle n| = |n\rangle \langle n| = \hat{\Pi}_n. \quad (2.17)$$

The POVM in Eq. (2.15) coincides with the one associated with the standard number measurement, and therefore yields the same probabilities. However, the state update is completely different from the projective one. This makes explicit that the POVM elements determine the outcome statistics, whereas the detection operators determine the state transformation.

The definition above extends naturally to continuous outcome spaces, where sums are replaced by operator-valued measures. For the present purposes, the discrete version is sufficient and already contains the essential ideas.

The relation between generalized measurements and ordinary projective measurements is clarified by the Naimark theorem [26]. First, we have to discuss about the ancilla. It is an auxiliary quantum system introduced in addition to the system of interest. It is typically used to model additional degrees of freedom involved in a physical process, such as a probe, a measurement apparatus, or an effective environment, and it allows one to describe indirect measurements and more general state transformations on an enlarged Hilbert space. Consider a system S coupled to an ancilla A , prepared in a reference state $|\omega_A\rangle$. Let the composite evolve unitarily, and let a projective measurement $\{\hat{P}_x\}$ be performed on the ancilla. Starting from the factorized state

$$\hat{\rho}_{SA} = \hat{\rho}_S \otimes |\omega_A\rangle \langle \omega_A|, \quad (2.18)$$

the probability of the outcome x can be written as

$$p_x = \text{Tr}_S \left[\hat{\rho}_S \hat{\Pi}_x \right], \quad (2.19)$$

where

$$\hat{\Pi}_x = \text{Tr}_A \left[(\mathbb{1}_S \otimes |\omega_A\rangle \langle \omega_A|) \hat{U}^\dagger (\mathbb{1}_S \otimes \hat{P}_x) \hat{U} \right]. \quad (2.20)$$

The operators $\hat{\Pi}_x$ are positive and satisfy completeness, therefore they define a POVM on $\mathcal{H}_S$.

The Naimark theorem states that the converse is also true: every POVM on $\mathcal{H}_S$ can be realized as a projective measurement on a larger Hilbert space, obtained by introducing suitable ancillary degrees of freedom and a unitary interaction. Generalized measurements are the effective description viewed as ordinary projective measurements performed indirectly on an enlarged system.

(METTERE IMMAGINE STILE PARIS - SCHEMA MISURA GENERALIZZATA)

2.1.2 Quantum maps and completely positive transformations

As we have seen before, the interaction with unobserved degrees of freedom, as well as the possibility of conditioning on measurement outcomes, leads to transformations that need not preserve purity and need not be reversible. This motivates the introduction of quantum maps, namely linear maps acting on statistical operators [25].

A quantum map is a rule that associates to each input state $\hat{\rho}$ an output operator $\Phi[\hat{\rho}]$. In the physical setting of interest, this rule must send density operators into density operators, so that positivity and trace are preserved. Moreover, since statistical mixtures must evolve consistently, the map has to be linear. In this way, quantum maps provide the natural generalization of the unitary transformation law of closed quantum systems. The unitary case is recovered when $\Phi[\hat{\rho}] = \hat{U} \hat{\rho} \hat{U}^\dagger$. In general, however, open-system transformations are not of this form: they may decrease coherences, transform pure states into mixed states, and describe irreversible processes. This is precisely the kind of behavior that cannot be captured within the closed-system framework alone. A simple example makes the difference clear. Consider a qubit and define the map

$$\Phi_p[\hat{\rho}] = (1 - p)\hat{\rho} + p \sigma_z \hat{\rho} \sigma_z, \quad 0 \leq p \leq 1, \quad (2.21)$$

called the phase-flip channel. This map is trace preserving and sends states into states. However, it is not unitary unless $p = 0$ or $p = 1$. In the computational basis $\{|0\rangle, |1\rangle\}$ and for $\rho = |+\rangle\langle +|$ one has

$$\Phi_p[\hat{\rho}_+] = \frac{1}{2} \begin{pmatrix} 1 & 1 - 2p \\ 1 - 2p & 1 \end{pmatrix}. \quad (2.22)$$

For $p = 1/2$ the output state is maximally mixed, and therefore a pure state is mapped into a mixed state. This is impossible under unitary evolution on the system Hilbert space alone.

The physically relevant class of transformations is selected by a few basic requirements. First, the map must be linear, in order to preserve the convex structure of statistical

mixtures. Second, it must preserve positivity and trace, so that density operators are mapped into density operators. Most importantly, it must be completely positive. This means that the extension of the map to any larger Hilbert space, $\Phi \otimes \mathbb{1}_A$ must still preserve positivity for every ancillary system A . This requirement is essential because the system of interest may be correlated, or even entangled, with additional degrees of freedom. Positivity on the system alone is thus not sufficient.

A quantum physical transformation is thus described by a completely positive trace-preserving map, usually abbreviated as CPTP map. More generally, if one considers a selective transformation associated with a specific measurement outcome, the trace need not be preserved. In that case one deals with a completely positive trace-non-increasing map.

2.1.3 Stinespring theorem and Kraus' operator-sum representation

The general structure of quantum maps is fixed by two important theorems. In finite dimension, this result states that any CPTP map can be represented in two equivalent ways: either as a reduced unitary evolution on a larger Hilbert space [27], or through an operator-sum representation [28].

Let S be the system of interest and E an ancilla or effective environment, prepared in a reference state $|\omega_E\rangle\langle\omega_E|$. Let the composite system evolve under a unitary operator $\hat{U}$ on $\mathcal{H}_S \otimes \mathcal{H}_E$. The reduced transformation of the system is then

$$\Phi[\hat{\rho}] = \text{Tr}_E \left[\hat{U} (\hat{\rho} \otimes |\omega_E\rangle\langle\omega_E|) \hat{U}^\dagger \right]. \quad (2.23)$$

This is the Stinespring form of the map. It shows that every admissible reduced dynamics may be viewed as ordinary unitary evolution on an enlarged Hilbert space followed by a partial trace.

To obtain the operator-sum representation, let $\{|k_E\rangle\}$ be an orthonormal basis of $\mathcal{H}_E$, and define the operators

$$\hat{K}_j = \langle k_E | \hat{U} | \omega_E \rangle. \quad (2.24)$$

These operators act on $\mathcal{H}_S$. By expanding the partial trace in the basis $\{|k_E\rangle\}$, one finds

$$\Phi[\hat{\rho}] = \sum_j \hat{K}_j \hat{\rho} \hat{K}_j^\dagger \quad (2.25)$$

they also satisfy the normalization condition. Equation (2.25) is called the operator-sum representation or Kraus representation of the map.

The converse statement also holds: given any family of operators $\{\hat{K}_j\}$ satisfying Eq. (2.26), the map in Eq. (2.25) is completely positive and trace preserving. In this way, the two descriptions are equivalent.

Unitary evolution is recovered as a special case. If there is only one Kraus operator, $\hat{K}_1 = \hat{U}$, then Eq. (2.25) reduces to the usual unitary evolution of the generic density operator.

This formalism also unifies dynamics and measurement. In the measurement setting, the Kraus operators are precisely the measurement operators introduced in the previous subsection. A selective outcome corresponds to a trace-non-increasing completely positive map, while the sum over all outcomes yields a CPTP map. For this reason, the Kraus representation provides the natural language for both reduced dynamics and generalized measurements.

It is worth remarking that the Kraus representation is not unique. Different sets of Kraus operators may describe the same map. This non-uniqueness reflects the fact that the same reduced transformation may arise from different dilations, or, equivalently, from different choices of basis in the ancillary Hilbert space.

The Kraus-Stinespring theorem is the key structural result needed in the following. It shows that the evolution of an open quantum system may always be described either in terms of a unitary interaction with additional degrees of freedom or, at the reduced level, by a completely positive map acting on the state of the system alone. This is precisely the framework in which master equations and continuously monitored dynamics will be formulated.

Conclusions

...

Appendix

...

Bibliography

- [1] Max Planck. “Ueber das Gesetz der Energieverteilung im Normalspectrum”. In: *Annalen der Physik* 309.3 (1901), pp. 553–563. DOI: [10.1002/andp.19013090310](https://doi.org/10.1002/andp.19013090310).
- [2] Albert Einstein. “Über einen die Erzeugung und Verwandlung des Lichtes betreffenden heuristischen Gesichtspunkt”. In: *Annalen der Physik* 322.6 (1905), pp. 132–148. DOI: [10.1002/andp.19053220607](https://doi.org/10.1002/andp.19053220607).
- [3] Niels Bohr. “On the Constitution of Atoms and Molecules”. In: *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*. 6th ser. 26.151 (1913), pp. 1–25. DOI: [10.1080/14786441308634955](https://doi.org/10.1080/14786441308634955).
- [4] Werner Heisenberg. “Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen”. In: *Zeitschrift für Physik* 33.1 (1925), pp. 879–893. DOI: [10.1007/BF01328377](https://doi.org/10.1007/BF01328377).
- [5] Max Born and Pascual Jordan. “Zur Quantenmechanik”. In: *Zeitschrift für Physik* 34.1 (1925), pp. 858–888. DOI: [10.1007/BF01328531](https://doi.org/10.1007/BF01328531).
- [6] Erwin Schrödinger. “Quantisierung als Eigenwertproblem”. In: *Annalen der Physik* 384.4 (1926), pp. 361–376. DOI: [10.1002/andp.19263840404](https://doi.org/10.1002/andp.19263840404).
- [7] Max Born. “Zur Quantenmechanik der Stoßvorgänge”. In: *Zeitschrift für Physik* 37.12 (1926), pp. 863–867. DOI: [10.1007/BF01397477](https://doi.org/10.1007/BF01397477).
- [8] John von Neumann. *Mathematical Foundations of Quantum Mechanics*. Translated by Robert T. Beyer. Princeton, NJ: Princeton University Press, 1955.
- [9] P. A. M. Dirac. *The Principles of Quantum Mechanics*. Oxford: Oxford University Press, 1930.
- [10] J. J. Sakurai and Jim Napolitano. *Modern Quantum Mechanics*. Cambridge: Cambridge University Press, 2017. ISBN: 9781108422413.
- [11] Michael Reed and Barry Simon. *Methods of Modern Mathematical Physics, Vol. 1: Functional Analysis*. Revised and Enlarged Edition. New York: Academic Press, 1980. ISBN: 9780125850506.
- [12] M. H. Stone. “On one-parameter unitary groups in Hilbert space”. In: *Annals of Mathematics*. Second Series 33.3 (1932), pp. 643–648. DOI: [10.2307/1968538](https://doi.org/10.2307/1968538).
- [13] Freeman J. Dyson. “The Radiation Theories of Tomonaga, Schwinger, and Feynman”. In: *Physical Review* 75.3 (1949), pp. 486–502. DOI: [10.1103/PhysRev.75.486](https://doi.org/10.1103/PhysRev.75.486).

- [14] Micha Horodecki, Pawe Horodecki, and Ryszard Horodecki. “Separability of Mixed States: Necessary and Sufficient Conditions”. In: *Physics Letters A* 223.1–2 (1996), pp. 1–8. DOI: [10.1016/S0375-9601\(96\)00706-2](https://doi.org/10.1016/S0375-9601(96)00706-2).
- [15] Asher Peres. “Separability Criterion for Density Matrices”. In: *Physical Review Letters* 77.8 (1996), pp. 1413–1415. DOI: [10.1103/PhysRevLett.77.1413](https://doi.org/10.1103/PhysRevLett.77.1413).
- [16] William K. Wootters. “Entanglement of Formation of an Arbitrary State of Two Qubits”. In: *Physical Review Letters* 80.10 (1998), pp. 2245–2248. DOI: [10.1103/PhysRevLett.80.2245](https://doi.org/10.1103/PhysRevLett.80.2245). arXiv: [quant-ph/9709029](https://arxiv.org/abs/quant-ph/9709029) [quant-ph].
- [17] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. 10th Anniversary. Cambridge: Cambridge University Press, 2010. ISBN: 9781107002173. DOI: [10.1017/CB09780511976667](https://doi.org/10.1017/CB09780511976667).
- [18] Ryszard Horodecki et al. “Quantum entanglement”. In: *Reviews of Modern Physics* 81.2 (2009), pp. 865–942. DOI: [10.1103/RevModPhys.81.865](https://doi.org/10.1103/RevModPhys.81.865).
- [19] Otfried Gühne and Géza Tóth. “Entanglement detection”. In: *Physics Reports* 474.1–6 (2009), pp. 1–75. DOI: [10.1016/j.physrep.2009.02.004](https://doi.org/10.1016/j.physrep.2009.02.004).
- [20] Masahiro Kitagawa and Masahito Ueda. “Squeezed Spin States”. In: *Physical Review A* 47.6 (1993), pp. 5138–5143. DOI: [10.1103/PhysRevA.47.5138](https://doi.org/10.1103/PhysRevA.47.5138).
- [21] D. J. Wineland et al. “Squeezed Atomic States and Projection Noise in Spectroscopy”. In: *Physical Review A* 50.1 (1994), pp. 67–88. DOI: [10.1103/PhysRevA.50.67](https://doi.org/10.1103/PhysRevA.50.67).
- [22] Jian Ma et al. “Quantum spin squeezing”. In: *Physics Reports* 509.2 (2011), pp. 89–165. ISSN: 0370-1573. DOI: <https://doi.org/10.1016/j.physrep.2011.08.003>. URL: <https://www.sciencedirect.com/science/article/pii/S0370157311002201>.
- [23] A. Sørensen and K. Mølmer. *Entanglement and extreme spin squeezing*. Aug. 2000. arXiv: [quant-ph/0011035](https://arxiv.org/abs/quant-ph/0011035).
- [24] Xiaoguang Wang and Barry C. Sanders. “Spin Squeezing and Pairwise Entanglement for Symmetric Multiqubit States”. In: *Physical Review A* 68.1 (2003), p. 012101. DOI: [10.1103/PhysRevA.68.012101](https://doi.org/10.1103/PhysRevA.68.012101).
- [25] Bassano Vacchini. *Open Quantum Systems*. Graduate Texts in Physics. Springer Nature Switzerland AG, 2024. ISBN: 978-3-031-58217-2. DOI: [10.1007/978-3-031-58218-9](https://doi.org/10.1007/978-3-031-58218-9).
- [26] M. G. A. Paris. “The modern tools of quantum mechanics: A tutorial on quantum states, measurements, and operations”. In: *The European Physical Journal Special Topics* 203 (2012), pp. 61–86. DOI: [10.1140/epjst/e2012-01535-1](https://doi.org/10.1140/epjst/e2012-01535-1).
- [27] W. Forrest Stinespring. “Positive Functions on C*-Algebras”. In: *Proceedings of the American Mathematical Society* 6.2 (1955), pp. 211–216. DOI: [10.1090/S0002-9939-1955-0069403-4](https://doi.org/10.1090/S0002-9939-1955-0069403-4).
- [28] K. Kraus. “General State Changes in Quantum Theory”. In: *Annals of Physics* 64.2 (1971), pp. 311–335. DOI: [10.1016/0003-4916\(71\)90108-4](https://doi.org/10.1016/0003-4916(71)90108-4).